P2H2P PCM Wellbore Storage

Mathematical Model of Power-to-Heat-to-Power Conversion
with Phase-Change Material in Repurposed Wells

Thermodynamic Cycles, Borehole Resistance Network,
Melt-Front Formulations, Sizing and Performance Indices

Model version 0.5 · last updated 2026-09-12 BRT

Craft & Hawkins Department of Petroleum Engineering
Louisiana State University

September 12, 2026

Abstract

This document states the governing equations of THUMS — thermal storage in repurposed wells with phase-change material — as the code actually implements them. It is a companion to the master-architecture report of the DBHE plant and follows the same convention: the model is written *against the source rather than from memory*, and every equation carries the function that evaluates it.

That convention matters more here than usual. Until this document, the THUMS equations existed nowhere except as Python. Four Colab notebooks and a conference paper were produced without a written model statement, and the principal defect corrected in v0.2 — the fluid side giving up heat through a finned surface of 0.4924 m while the melt front absorbed it through a bare cylinder of 0.1324 m — is of exactly the kind that a document placing both surfaces on the same page would have exposed immediately.

Section 1 fixes notation and states the modelling hypotheses. Sections 2–5 give the model proper. Section 7 is not an appendix of caveats but part of the specification: it lists the assumptions that are known to be violated, the two places where the same physical quantity is given two different formulas, and the terms that are absent altogether. A reader reproducing these results needs that section as much as the equations.

Two melt-front formulations are given. Section 4.2 states the closed-form Stefan solve used through v0.1 and the conference paper; Section 4.3 states the energy-balance formulation that replaced it. Both remain in the code and are selected by `Case.front`, so the difference between them is measurable rather than arguable.

Changes in version 0.3. Two corrections, both now the default. The tube-wall and fin conductivity used during charging was corrected from the PCM liquid value to steel (§7.4); and the fin metal is now excluded from the melted PCM cross-section (§4.4).

Changes in version 0.4. The melt front now carries *enthalpy* rather than melted area (§4.5), so the PCM may superheat above T_m once fully molten and subcool below it once fully solid. Formulation B previously had nowhere to put heat delivered to a saturated segment and discarded it — on discharge the discarded heat reached 148 % of the heat delivered. Three things follow: no heat is clipped, the local melt fraction is confined to $[0, 1]$ by construction, and desuperheating and subcooling are recovered during discharge. The well count is now set by a *capacity* criterion (§5.2) rather than by the melted-volume constraint of v0.3.

The consequence worth noting is not the headline number but the *uniformity*: sensible heat is self-levelling, because a segment that finishes melting superheats, its driving temperature

difference collapses, and the heat passes downstream. The spread in local melt fraction along the well falls from 1.321 to 0.078.

Changes in version 0.4a. No physics. Two reporting corrections, both of which changed a conclusion. Section 4.5 is rewritten at length as a teaching treatment of Formulation C — the choice of state variable, the enthalpy curve, the branchwise-exact integration and its verification, and the failures that motivated each. Section 5.2 adds a reporting rule: N_{capacity} is a lower bound and must not be read across a parameter sweep on its own, because N_{wells} is a maximum and the binding criterion changes. Section 7.1 corrects the geometric limit on the melt front from the borehole wall to the *cell* radius, Eq. (81) — the old check was a factor 2.9 too permissive in δ and could never fire. Under Formulation C the corrected limit cannot be violated, but it is being *saturated* over 99 % of the well, which now bears on the fin count and is the leading open item.

Changes in version 0.5. Section 6 is new and addresses cyclic operation. Everything before it is a *first-cycle* result from a cold start, and the cycle does not return to that state. At cyclic steady state the model has $\eta_{\text{storage}} \equiv 1$ exactly, because the state returns to itself and there is no loss path; two consequences follow. The assumed loss surplus λ becomes unattainable — the CSS charge pins at $1/(1+\lambda)$ of the requirement for every well count — and even at $\lambda = 0$ the rate criterion goes degenerate, fixing the flow ratio rather than N_{wells}. The resolution is to specify the hardware and *simulate* rather than solve: N_{wells} from latent heat alone, both flows pinned by their glides, march to CSS, and report the deviation from target. At the design point the field delivers 94.26 % of the 1 MW target, and the shortfall is a *discharge rate* limit rather than an inventory limit.

Cycling also exposed a defect in the existing single-cycle path (§6.4): the melt state was handed to the discharge without being mirrored, although the flow reverses and the cascade is mirrored with it, moving the enthalpy datum by up to 48.9 K and reporting PCM hotter than any fluid that had touched it. N_{wells} is unaffected; the discharge-side indices move by up to 2 %. Section 4.5.5 derives the segment equation in full — the two control volumes, the quasi-steady fluid approximation and what it costs, and why the closing conductance is $\dot{m}c_p(1-e^{-\text{NTU}})/\Delta z$ rather than $U_i P_i$, a difference reaching 33 % here. A ninth hypothesis, **H9**, is stated explicitly for the first time: perfect azimuthal insulation between the cascade layers of a hairpin's two legs, which at the wellhead are 48.9 K apart. It is assumed perfect for now and quantified in §7.2, where the neglected leak is of order 25 % of the useful duty.

Contents

1 Scope, hypotheses and notation

THUMS stores electrical energy as latent heat in a phase-change material packed into the annulus of a repurposed oil-and-gas well. A high-temperature heat pump melts the PCM during charging; an organic Rankine cycle recovers the energy during discharging. The storage medium is a tri-modal PCM melting at 150 °C; the heat-transfer fluid is pressurised water; the borehole contains longitudinally finned hairpin tubes.

The model answers two questions: *how many wells does a given power block need*, and *what round-trip efficiency does the resulting plant achieve*. Everything below serves those two numbers.

Headline results, version 0.4

	v0.1 published	marched, old k_w	**v0.4**
N_{wells}	24.123	22.431	**11.573**
binding constraint	—	charge rate	**capacity**
ε_{PCM}	0.5200	0.5402	**0.9992**
local ε, min–max	—	0.46–1.78	**0.922–1.000**
η_{RTE}	0.43027	0.42955	0.41585
η^0_{RTE}	0.43484	0.43484	0.43484
COP	2.95633	2.95633	2.95633
η_{ORC}	0.23685	0.23685	0.23685
η_{storage}	—	0.953	0.9364
closure error	3–54 %	10^{-16}	10^{-16}

The sizing moves a long way and the efficiency hardly at all: N_{wells} falls by 52 % while η_{RTE} moves by about 3 %. η^0_{RTE} is identical to twelve significant figures throughout — the cycles are untouched by any of this. What the storage model determines is how much hardware the cycle needs, not how efficiently it runs.

1.1 The structure of the model

The calculation is a chain, not a coupled system. Each stage consumes the previous stage's output and nothing flows backwards:

1. the ORC and heat-pump cycles fix η_{ORC} and COP from CoolProp property calls alone (§2.1);
2. an energy budget converts the specified electrical output into a charging heat duty and a secondary-fluid mass flow (§2.2);
3. a resistance network gives the local overall coefficient U_i along the tube (§3);
4. an effectiveness–NTU march integrates the fluid temperature down the tube, segment by segment (§4.1);
5. a melt-front model converts the delivered heat into a melted volume (§4.2 or §4.3);
6. the well count is chosen so that the field satisfies its constraints, and the performance indices follow (§5).

The consequence worth stating explicitly is that the cycles are decoupled from the storage. η_{ORC} and COP do not depend on anything the borehole does. No change to the storage model can move them, which is why the ideal round-trip efficiency η^0_{RTE} of Eq. (71) is invariant across every formulation in this document.

1.2 Modelling hypotheses

The following are assumed throughout. Those known to be violated in the operating regime are revisited in §7.

H1 — Quasi-steady conduction. The melt layer carries no thermal mass. Its temperature profile is the steady cylindrical profile corresponding to the instantaneous front position. Time enters the model *only* through the front position. The hypothesis is controlled by the Stefan number

$$\mathrm{Ste} = \frac{c_{p,m}\,\Delta T}{h_m}, \tag{1}$$

and degrades as it grows; at the design point $\mathrm{Ste} = 0.0474$.

H2 — No axial conduction. Neither the tube wall nor the PCM conducts along the well. Segments are coupled only through the fluid stream. This one is safe by a wide margin: the axial diffusion time over a segment is $\Delta z^2/\alpha_l = 5.4\mathrm{e}9\,\mathrm{s}$ against a 3.6e4 s window (§4.5.5).

H3 — Concentric annular front. The melt region around each tube is a cylindrical annulus of uniform thickness δ, so that the melted cross-section is $\pi[(r_e+\delta)^2 - r_e^2]$. Fronts from neighbouring tubes do not interact and do not merge. *Under Formulation C this cannot be violated, but it is saturated over the whole well — see §7.1.*

H4 — Conduction only in the melt. No natural convection in the liquid PCM. This is conservative early and increasingly wrong as δ grows.

H5 — Enthalpy state, no radial resolution. *Revised in v0.4.* The PCM carries latent *and* sensible energy, lumped into one enthalpy per segment (§4.5): fully molten PCM superheats, fully solid PCM subcools. What is still absent is any radial resolution of temperature *within* a phase — each segment has a single PCM temperature, so the thermal gradient through the melt layer is quasi-steady (H1) rather than resolved.

H6 — Adiabatic borehole wall. No heat exchange with the formation in either direction, and no geothermal gradient in the energy balance.

H7 — Incompressible, single-phase secondary fluid. Pressurised water at constant P; no buoyancy head in the vertical loop. The fluid is also taken as *quasi-steady*: $\partial T/\partial t$ is dropped from its energy equation, which costs an unresolved startup transient of about 38 min in each half-cycle (§4.5.5).

H8 — Multilayer PCM. The well is divided into N_{lay} axial layers of distinct melting temperature, so that the PCM melting point follows the fluid glide down the well. Layer boundaries are equally spaced along the *developed* tube length L_{tube}, not the depth — so a hairpin's down leg and return leg belong to *different* layers at the same depth. See H9.

H9 — Perfect azimuthal insulation. *New in v0.4a, and the weakest assumption in the set.* Each tube leg owns an equivalent PCM cell of radius r_{cell} (Eq. (81)) whose outer boundary is adiabatic: no heat crosses between the cells of neighbouring legs at the same depth. Because H8 cascades along the developed length, those neighbours are at *different* melting temperatures — up to 48.9 K apart at the wellhead, falling to zero at the well bottom where the two legs meet. The assumption therefore requires a physical partition between the down-leg and return-leg PCM *and* that the partition be insulating. Neither is designed, costed, or modelled. A slab estimate puts the neglected leak at $\sim 22\,\mathrm{W/m}$ against a useful duty of $\sim 83\,\mathrm{W/m}$ at the wellhead — of order 25 %, and acting to erase precisely the temperature separation the cascade exists to create. Held as an idealisation pending evaluation; §7.2.

1.3 Geometry

One borehole of diameter D_{well} and depth L_{well} contains n_t hairpin tubes. A hairpin runs down and back, so its developed length is

$$L_{\text{tube}} \;=\; 2\,L_{\text{well}}, \tag{2}$$

and the number of tube *legs* in the borehole is $2n_t$. This factor-of-two bookkeeping is a recurring source of error: per-borehole quantities multiply the per-tube result by n_t, not by $2n_t$, because Eq. (2) has already counted both legs.

The distinction between the developed coordinate $s \in [0, L_{\text{tube}}]$ and the depth d matters wherever a profile is plotted or a layer assigned:

$$d(s) = \begin{cases} s, & s \le L_{\text{well}} \quad \text{(down leg)}, \\ L_{\text{tube}} - s, & s > L_{\text{well}} \quad \text{(return leg)}. \end{cases} \tag{3}$$

Every depth therefore carries *two* legs of the same hairpin, at developed positions s and $L_{\text{tube}} - s$, and by H8 those two are in different PCM layers. H9 assumes they do not exchange heat.

Each leg carries n_f longitudinal fins of thickness t_f and radial length L_f. The PCM volume available in one borehole is the borehole volume less the tubes and their fins,

$$V_{\text{well}} \;=\; \frac{\pi D_{\text{well}}^2}{4}\,L_{\text{well}} \;-\; 2n_t\Big(\pi r_e^2 + n_f t_f L_f\Big)L_{\text{well}}. \tag{4}$$

Equation (4) deserves a second look, because it is the source of a reporting trap taken up in §5.2: fin metal displaces PCM, so *any* change to the fins moves V_{well} and hence the storage capacity of a borehole, entirely independently of what the fins do thermally.

Source: `thums_core/config.py`, class `Geometry`.

1.4 Nomenclature

symbol	meaning	unit
A_{melt}	melted cross-section per unit tube length	m^2
A_{flow}	tube flow area, πr_i^2	m^2
d	depth below surface	m
s	developed tube coordinate	m
A_{avail}	PCM cross-section owned by one segment	m^2
C_s, C_l	sensible capacity per unit tube length	J/m/K
E'	PCM enthalpy per unit tube length	J/m
E'_{lat}	latent capacity per unit tube length	J/m
K	segment conductance per unit tube length	W/m/K
c_p	specific heat capacity	J/kg/K
D_{well}	borehole diameter	m
h_m	latent heat of fusion	J/kg
h_i	internal convective coefficient	$\text{W/m}^2\text{/K}$
h_e	melt-layer conductance, outer side	$\text{W/m}^2\text{/K}$
k_m	PCM thermal conductivity	W/m/K
k_w	tube-wall and fin conductivity	W/m/K
L_f, t_f, n_f	fin length, thickness, count	m, m, –

symbol	meaning	unit
L_{well}, L_{tube}	well depth, developed tube length	m
$\dot{m}$	mass flow rate	kg/s
n_t	hairpin tubes per borehole	–
N_{lay}	PCM layers along the well	–
N_{wells}	number of boreholes	–
NTU	number of transfer units, per segment	–
q'	heat rate per unit tube length	W/m
r_i, r_e	tube inner and outer radius	m
r_{cell}	equivalent cell radius of one tube leg	m
R'	thermal resistance referred to inner area	$\text{m}^2\text{K/W}$
$R''_{f,i}$	internal fouling resistance	$\text{m}^2\text{K/W}$
t_{ch}, t_{dc}	charging, discharging duration	h
T_m	PCM melting temperature	K
T_{pcm}	PCM temperature, recovered from E'	K
T_{r_e}	tube outer-surface temperature	K
U_i	overall coefficient on inner area	$\text{W/m}^2\text{/K}$
V_{well}, V_{melt}	PCM volume, melted volume	m^3
δ	melt-layer thickness	m
δ_{merge}	thickness at which neighbouring fronts touch	m
Δz	segment length	m
ε_{PCM}	melted fraction of the PCM inventory	–
$\varepsilon_{\text{local}}$	melted fraction of one segment's PCM	–
η_f, η_o	fin and overall surface efficiency	–
λ	assumed storage-loss surplus	–
ρ_m	PCM density	kg/m^3

Subscripts s and l denote the solid and liquid PCM phases. Subscripts ch and dc denote charging and discharging. A prime on a resistance denotes referral to the inner tube area, and a prime on a heat rate denotes per unit tube length.

2 Thermodynamic cycles and the energy budget

2.1 State points

The plant contains two cycles. During charging a two-stage high-temperature heat pump with two regenerators lifts heat from a source at T_{source} to the PCM melting temperature. During discharging a two-stage organic Rankine cycle rejects to a sink at T_{sink}. Both use cyclopentane; the secondary fluid circulating through the boreholes is pressurised water.

Every state point is fixed by a specified approach temperature rather than by a component model. Writing T_m for the PCM melting point,

$$
\begin{aligned}
T_{1e} &= T_{\text{sink}} + \Delta T_{E,\text{sink}}, & T_{2d} &= T_m - \Delta T_{m,2D}, && (5)\\
T_{3e} &= T_{2d} - \Delta T_{2D,3E}, & T_{3d} &= T_{2d} - \Delta T_{2D,3D}, && (6)\\
T_{4c} &= T_m + \Delta T_{4C,M}, & T_{4a} &= T_{\text{source}} - \Delta T_{3A,4A}, && (7)\\
T_{13h} &= T_{\text{source}} - \Delta T_{3A,13H}, & T_{3c} &= T_{4c}, && (8)\\
T_{2c} &= T_{3c} - \Delta T_{3C,2C}, & T_{2h} &= T_{3c} + \Delta T_{2H,3C}. && (9)
\end{aligned}
$$

The secondary-fluid glide across the borehole field, $\Delta T_{3C,2C}$, is the principal design variable of the study. The discharge-side glide is tied to it,

$$\Delta T_{2D,3D} \;=\; \Delta T_{3C,2C}, \tag{10}$$

by construction. In v0.1 these were two independent variables carrying a comment that they must be equal; a comment is not a constraint, and Eq. (10) now enforces it.

Given the state points, the cycle efficiencies

$$\eta_{\text{ORC}} \;=\; \frac{w_{\text{net}}}{q_{\text{in}}}, \qquad \text{COP} \;=\; \frac{q_{\text{out}}}{w_{\text{comp}}} \tag{11}$$

follow from enthalpies evaluated by CoolProp at those points.

Source: `_legacy_physics.double_stage_rankine`, `two_stage_htheatpump_2regs`; assembled in `system._cycles`.

Caveat, stated here because it bounds every efficiency in this document. The expansion and compression steps are evaluated between the prescribed state points without isentropic efficiencies. η_{ORC} and COP are therefore idealised, and the round-trip efficiencies of §5.3 inherit that idealisation. Only the electrical and mechanical efficiencies η_T, η_G, η_C, η_H of Eq. (12) appear. This is a property of the model as built, not a simplification introduced by this document.

2.2 Energy budget

The chain runs backwards from the specified net electrical output $\dot{W}_{\text{el,out}}$ of the power block:

$$
\begin{aligned}
\dot{W}_T &= \frac{\dot{W}_{\text{el,out}}}{\eta_T\,\eta_G}, & \dot{Q}_{\text{in,ORC}} &= \frac{\dot{W}_T}{\eta_{\text{ORC}}},\\
\Delta E_{\text{in,ORC}} &= \dot{Q}_{\text{in,ORC}}\,t_{\text{dc}}, & \Delta E_{\text{out,HP}} &= \Delta E_{\text{in,ORC}}\,(1+\lambda),\\
\dot{Q}_{\text{out,HP}} &= \frac{\Delta E_{\text{out,HP}}}{t_{\text{ch}}}, & \dot{W}_{C,\text{HP}} &= \frac{\dot{Q}_{\text{out,HP}}}{\text{COP}},\\
\dot{W}_{\text{el,in}} &= \frac{\dot{W}_{C,\text{HP}}}{\eta_C\,\eta_H}.
\end{aligned} \tag{12}
$$

The charging mass flow of secondary fluid follows from the duty and the glide,

$$\dot{m}_w^{\text{ch}} \;=\; \frac{\dot{Q}_{\text{out,HP}}}{c_{p,w}\,\Delta T_{3C,2C}}, \qquad c_{p,w} = c_p\big(\tfrac{1}{2}(T_{3c}+T_{2c}),\;P\big)\,, \tag{13}$$

and is divided equally among all tubes of all wells,

$$\dot{m}_1^{\text{ch}} \;=\; \frac{\dot{m}_w^{\text{ch}}}{N_{\text{wells}}\,n_t}. \tag{14}$$

Note for §5.2: Eq. (13) *pins* the charging flow to the specified glide. It is therefore not free, and this is what decides which side of the cycle can size the well field.

2.3 The storage-loss surplus λ

The factor $(1+\lambda)$ in Eq. (12) is the one place where the budget anticipates a loss it does not compute. It asserts that charging must deposit λ more energy than discharging will recover, and it is set to $\lambda = 0.05$ by fiat.

In v0.1 this could not be checked, because the discharge model began from a bare tube in fully solid PCM irrespective of what charging had produced (§4.2). With the melt state carried across the cycle (§4.3) the recovered fraction becomes an output,

$$\eta_{\text{storage}} = \frac{\Delta E_{\text{discharged}}}{\Delta E_{\text{stored}}}, \tag{15}$$

and can be compared with the $1/(1+\lambda) = 0.952$ that Eq. (12) presumes. At the design point with the legacy wall conductivity $\eta_{\text{storage}} = 0.953$, and the assumption survives. With the correct steel conductivity it falls to 0.9364, and it does not: the field banks about 7 % more than the ORC withdraws, against the 5 % assumed. Making λ an output rather than an input remains open work.

3 The borehole resistance network

At a given axial station the heat leaving the secondary fluid crosses, in series: the internal convective film, the fouling layer, the tube wall, and the melt layer standing on a finned outer surface. All four are referred to the inner tube area, so that a single coefficient U_i drives the segment march of §4.1.

3.1 Internal convection

With $D_i = 2r_i$ and $\dot{m}$ the flow in one tube,

$$\text{Re} = \frac{4\dot{m}}{\pi D_i\,\mu}, \qquad \text{Pr} = \frac{c_p\mu}{k}. \tag{16}$$

The Nusselt number is the Gnielinski correlation in turbulent flow and a fixed fully-developed value in laminar flow:

$$\text{Nu} = \begin{cases} \text{Nu}_{\text{lam}}, & \text{Re} < 2300, \\ \dfrac{(f_G/8)(\text{Re}-1000)\,\text{Pr}}{1+12.7\sqrt{f_G/8}\,\left(\text{Pr}^{2/3}-1\right)}, & \text{Re} \geq 2300, \end{cases} \qquad f_G = \left(0.79\ln\text{Re} - 1.64\right)^{-2}, \tag{17}$$

and

$$h_i = \frac{\text{Nu}\,k}{D_i}. \tag{18}$$

The laminar constant differs between the two implementations: $\text{Nu}_{\text{lam}} = 4.36$ in `well_marched` (uniform wall flux) and 3.66 in the legacy `temperature_profile_melt` (uniform wall temperature). For a melting boundary the wall is closer to isothermal, so 3.66 is the better constant of the two; the discrepancy is recorded in §7 and is inconsequential at the design point, where the flow is strongly turbulent.

3.2 Wall, fouling and melt layer

The wall and fouling resistances, referred to the inner area, are

$$R'_{\text{wall}} = \frac{r_i}{k_w}\ln\frac{r_e}{r_i}, \qquad R'_f = R''_{f,i}, \qquad R'_{\text{conv}} = \frac{1}{h_i}. \tag{19}$$

The melt layer is treated as a steady cylindrical conduction shell between r_e and $r_e+\delta$. Expressed as a conductance on the tube outer surface,

$$h_e \;=\; \frac{k_m}{r_e\,\ln\!\big(1+\delta/r_e\big)}. \tag{20}$$

As $\delta \to 0$ the layer vanishes and $h_e \to \infty$; the code caps this at 10^9, which is the correct limit and not, as an earlier audit asserted, a defect. What *is* defective is that error paths also route into that branch, so a failed solve is reported as maximum heat transfer (§7).

Equation (20) carries an assumption that becomes important in §7.1: it presumes the *entire* azimuth $2\pi(r_e+\delta)$ is available for conduction at every radius. That holds only while the melt front is free of its neighbours.

3.3 Fin efficiency and the finned surface

The fins are straight longitudinal fins of rectangular profile, treated with the adiabatic-tip correction $L_c = L_f + t_f/2$:

$$m \;=\; \sqrt{\frac{2h_e}{k_w\,t_f}}, \qquad \eta_f \;=\; \frac{\tanh(mL_c)}{mL_c}. \tag{21}$$

With the fin surface and total outer surface per tube

$$A_f \;=\; L_{\text{tube}}\,\big(2L_f + t_f\big), \qquad A_t \;=\; 2L_{\text{tube}}\,\big(\pi r_e + n_f L_f\big), \tag{22}$$

the overall surface efficiency is

$$\eta_o \;=\; 1 - \frac{n_f A_f}{A_t}\,\big(1-\eta_f\big), \tag{23}$$

and the wetted perimeter of the finned tube is

$$P_T \;=\; 2\pi r_e + 2n_f L_f. \tag{24}$$

Equation (24) is the quantity at the centre of the v0.2 correction. At the design geometry $P_T = 0.4924\,\text{m}$, against a bare-tube perimeter $2\pi r_e = 0.1324\,\text{m}$ — a factor of 3.72. The legacy melt-front model of §4.2 uses the bare perimeter while the network above uses P_T, and nothing in v0.1 compared them.

How the fins enter, and how they do not. Equations (21)–(24) represent a fin as *additional surface at the tube*: more perimeter, discounted by an efficiency. That is the whole of the fin model. A fin is *not* represented as a radial conduction path carrying heat outward to PCM the tube cannot easily reach. The distinction is immaterial while the melt layer is thin and becomes material once the fronts approach one another, which is the situation analysed in §7.1.

Note also that k_w appears twice with different physical meanings: as the tube-wall conductivity in Eq. (19) and as the *fin* conductivity in Eq. (21). A single argument carries both. This is what allowed the PCM liquid conductivity to be passed in that slot for months without contradiction (§7.4).

3.4 Overall coefficient

The melt-layer resistance referred to the inner area carries the ratio of perimeters and the surface efficiency,

$$R'_{\text{outer}} \;=\; \frac{2\pi r_i}{P_T\,\eta_o\,h_e}, \tag{25}$$

and the overall coefficient is the inverse of the series sum,

$$\boxed{U_i \;=\; \Bigg[\; \underbrace{\tfrac{1}{h_i}}_{\text{convection}} \;+\; \underbrace{R''_{f,i}}_{\text{fouling}} \;+\; \underbrace{\tfrac{r_i}{k_w}\ln\tfrac{r_e}{r_i}}_{\text{wall}} \;+\; \underbrace{\tfrac{2\pi r_i}{P_T\eta_o h_e}}_{\text{melt layer}} \;\Bigg]^{-1}.} \tag{26}$$

Source: `_legacy_physics.compute_U_i`.

3.5 Outer-surface temperature

The legacy melt-front model is driven by the tube outer-surface temperature rather than by the heat flow. Splitting the network at r_e, with

$$R_1 \;=\; R''_{f,i} + \frac{1}{h_i} + \frac{r_i}{k_w}\ln\frac{r_e}{r_i}, \qquad R_2 \;=\; \frac{2\pi r_i}{P_T\, h_e\, \eta_o}, \tag{27}$$

and $T_{\text{avg}} = \frac{1}{2}(T_j + T_{j+1})$ the mean fluid temperature over the segment, a series voltage-divider gives

$$T_{r_e} \;=\; T_{\text{avg}} - \frac{R_1\big(T_{\text{avg}} - T_m\big)}{R_1 + R_2}. \tag{28}$$

Source: `_legacy_physics.compute_T_re`.

Equations (26) and (28) are solved together with the melt front in a fixed-point loop at each segment, since h_e depends on δ, δ depends on T_{r_e}, and T_{r_e} depends on h_e. The loop runs to a tolerance δ_{tol} or a maximum iteration count, warm-started from the previous segment.

Equation (28) is required *only* by the legacy front. The formulations of §4.3 and §4.5 advance the front from q' directly and never form T_{r_e}, which removes the fixed-point loop along with it.

4 Axial march and the melt front

4.1 Effectiveness–NTU segment march

The tube is divided into n_s segments of length $\Delta z = L_{\text{tube}}/n_s$. Nodes are placed at $z_j = j\Delta z$, $j = 0, \ldots, n_s$, and each node is assigned the melting temperature of the PCM layer containing it,

$$T_m^{(j)} \;=\; T_{m,\ell} \quad \text{for} \quad z_j \in \big[\, z_\ell,\, z_{\ell+1}\big), \qquad z_\ell = \frac{\ell}{N_{\text{lay}}}\, L_{\text{tube}}. \tag{29}$$

During charging the layer melting temperatures descend with the fluid glide, $T_{m,\ell} = T_m - \ell\,\Delta T_{3C,2C}/N_{\text{lay}}$; during discharging they ascend, $T^{\text{dc}}_{m,\ell} = T_m - \Delta T_{3C,2C} + (\ell+1)\Delta T_{3C,2C}/N_{\text{lay}}$.

Within a segment the PCM surface is isothermal at $T_m^{(j)}$, so the segment is a single-stream heat exchanger of effectiveness $1 - e^{-\text{NTU}}$:

$$\text{NTU}_j \;=\; \frac{2\pi r_i\, U_i^{(j)}\, \Delta z}{\dot m\, c_p}, \qquad T_{j+1} \;=\; T_m^{(j)} + \big(T_j - T_m^{(j)}\big)\, e^{-\text{NTU}_j}. \tag{30}$$

The heat given up by the fluid over the segment, per unit tube length, is

$$q'_j \;=\; \frac{\dot m\, c_p\, \big(T_j - T_{j+1}\big)}{\Delta z}, \tag{31}$$

positive when the fluid is melting PCM and negative when the PCM is freezing. The march is sequential: $T_0 = T_{\text{inlet}}$, and each segment's outlet is the next segment's inlet.

Source: `_legacy_physics.temperature_profile_melt`; `well_marched.march`.

The two formulations report different heat rates. Formulation A forms the tube heat rate as an *endpoint enthalpy difference* over the whole tube,

$$Q_{\mathrm{A}} \;=\; \dot m\,\bigl[h(T_0,P) - h(T_{n_s},P)\bigr], \tag{32}$$

evaluated by CoolProp, whereas Formulation B accumulates the segment contributions actually taken up by the PCM, $Q_{\mathrm{B}} = \sum_j q'_{\mathrm{eff},j}\,\Delta z$ (Eq. (38)).

Equation (32) is the better measure of what the *fluid* gave up, since it uses real enthalpies rather than the local c_p of Eq. (31). But it is not the heat the *front* absorbed, and in Formulation A nothing requires the two to agree — the front is advanced from T_{r_e} through Eq. (35), never from Q_{A}. That is the closure defect of §7 stated in one line.

4.2 Formulation A — closed-form Stefan front (v0.1)

This is the formulation behind the conference paper. It is selected by `Case(front="closed_form")` and is retained as a regression reference.

The front is obtained from a quasi-steady Stefan solution for a bare concentric cylinder, driven by the outer-surface temperature T_{r_e} of Eq. (28). With $\alpha_m = k_m/(\rho_m c_{p,m})$, the Fourier and phase-change numbers

$$\mathrm{Fo} \;=\; \frac{\alpha_m t}{r_e^2}, \qquad \mathrm{Ph} \;=\; \left|\frac{h_m}{c_{p,m}\bigl(T_{r_e}-T_m\bigr)}\right| \;=\; \frac{1}{\mathrm{Ste}}, \tag{33}$$

and the dimensionless front position $x = 1 + \delta/r_e$, the front solves

$$\boxed{\mathrm{Fo} \;=\; \mathrm{Ph}\left[\tfrac12 x^2\ln x - \tfrac14 x^2 + \tfrac14\right]} \tag{34}$$

which, on substituting Fo and Ph, is equivalently

$$\tfrac12 x^2\ln x - \tfrac14 x^2 + \tfrac14 \;=\; \frac{k_m\bigl(T_{r_e}-T_m\bigr)\,t}{\rho_m\,h_m\,r_e^2}. \tag{35}$$

Equation (35) is the exact quadrature of $\rho_m h_m\,\mathrm{d}(\pi s^2)/\mathrm{d}t = 2\pi k_m(T_{r_e}-T_m)/\ln(s/r_e)$, with $s = r_e + \delta$, and is solved for δ by safeguarded Newton–bisection on $[0, \delta_{\max}]$.

Source: `_legacy_physics.compute_delta2_fast`.

Two assumptions, both violated. Equation (35) is derived for a *bare* cylinder of perimeter $2\pi r_e$, whereas the fluid side gives up its heat through the finned perimeter P_T of Eq. (24). The two surfaces differ by 3.72, so the heat leaving the fluid and the latent heat absorbed by the front are not the same energy. Second, it assumes T_{r_e} has held constant since $t = 0$, whereas T_{r_e} is applied at its current value; measured at mid-well, $T_{r_e} - T_m$ runs from 0 to 8.62 K across the charging window.

These two errors act in opposite directions and are quantified in §7. Neither is a numerical defect: the solver returns the exact root of Eq. (35) to a residual of 10^{-15}. It is the premises that fail.

4.3 Formulation B — energy-balance front (v0.2)

The correction observes that of the two models — heat delivered, and front position — only one may be specified freely. The other follows from conservation. The front is therefore advanced by the heat the network actually delivers:

$$\boxed{\rho_m\, h_m\, \frac{\mathrm{d}A_{\mathrm{melt}}}{\mathrm{d}t} \;=\; q'(t).} \tag{36}$$

Integrated explicitly over the time levels t^n, with $\Delta t = t^{n+1} - t^n$,

$$A_{\mathrm{melt}}^{n+1} \;=\; \mathcal{P}\!\left[\, A_{\mathrm{melt}}^{n} + \frac{q'\,\Delta t}{\rho_m h_m} \,\right], \qquad \mathcal{P}[A] = \min\!\big(\max(A, 0),\, A_{\max}\big), \tag{37}$$

where the projection $\mathcal{P}$ enforces two physical bounds: PCM that is already solid cannot freeze further, and a segment cannot melt more PCM than it contains. The heat actually taken up is recovered from the projected state,

$$q'_{\mathrm{eff}} \;=\; \frac{\big(A_{\mathrm{melt}}^{n+1} - A_{\mathrm{melt}}^{n}\big)\rho_m h_m}{\Delta t}, \tag{38}$$

and it is q'_{eff}, not q', that is accumulated into the delivered energy and that sets the segment outlet temperature in Eq. (30). Accumulating q' instead would credit the fluid with heat no PCM supplied.

Three properties follow, none of which Formulation A possesses.

Closure is structural. $\int q'\,\mathrm{d}t = \rho_m h_m \Delta V_{\mathrm{melt}}$ holds identically rather than approximately. This is a consistency property, *not* evidence that the model is correct: Eqs. (37) and (38) are inverse operations, so a residual of 10^{-16} measures floating-point arithmetic and nothing else.

The fins enter the front automatically, because q' comes from the network of Eq. (26), which carries P_T and η_o.

T_{r_e} **need not be constant.** The state is marched, so a varying inlet temperature is handled correctly and Eq. (28) is not required.

4.4 The fins are metal, not PCM

A_{melt} is the quantity the latent balance conserves, so it must be PCM and nothing else. The annulus between r_e and $r_e+\delta$ is not all PCM: the fins occupy metal inside it, of cross-section $n_f t_f \min(\delta, L_f)$, which saturates once the front passes the fin tips. Hence

$$A_{\mathrm{melt}} \;=\; \pi\left[(r_e+\delta)^2 - r_e^2\right] \;-\; n_f\, t_f\, \min(\delta,\, L_f)\,. \tag{39}$$

Inverting Eq. (39) for δ is still closed form — the fin term only shifts the quadratic, and each branch is solved exactly:

$$\delta \le L_f: \qquad \pi\delta^2 + (2\pi r_e - n_f t_f)\,\delta - A_{\mathrm{melt}} = 0, \tag{40}$$

$$\delta > L_f: \qquad \pi\delta^2 + 2\pi r_e\delta - (A_{\mathrm{melt}} + n_f t_f L_f) = 0. \tag{41}$$

With $n_f = 0$ both collapse to $\delta = \sqrt{r_e^2 + A_{\mathrm{melt}}/\pi} - r_e$, the bare concentric annulus used before this correction.

Corrected in v0.3. Earlier versions treated the whole annulus as PCM. That overstated the melted volume by 11 % when the layer is thin and 6 % at the design point, and understated δ — hence the melt-layer conduction resistance of Eq. (20) — by up to 34 %. Because it inflated $\varepsilon_{\mathrm{PCM}}$ it also pushed $N_{\mathrm{inventory}}$ slightly *up*; correcting it moves the design point from 12.79 to 12.74 wells.

What remains unmodelled is the front *shape*. The fins enhance the heat transfer through η_o, but the melt region is still taken as a circular annulus (hypothesis H3); physically it bulges along the fins.

4.5 Formulation C — enthalpy state (v0.4)

This subsection is written at greater length than the rest of the document. Formulation C is the part of the model a new student is most likely to have to modify, and it is the part where the reasoning — not the final formula — is what has to be understood. Readers who only want the equations can take Eqs. (43)*,* (52) *and* (54) *and skip the discussion.*

4.5.1 The question a state variable answers

Every transient model must decide what it *remembers* from one instant to the next. That choice — the state variable — is not cosmetic. It decides which physical situations the model can represent at all, because anything the state cannot encode is a situation the model cannot be in.

The three formulations in this section are three answers to that one question, and they are best read as a sequence:

	state variable	what it can represent
A (v0.1)	δ, from a closed form	a melt front, at one prescribed T_{r_e}
B (v0.2)	A_{melt}, integrated	a melt front, driven by the actual heat flow
C (v0.4)	E', integrated	a melt front *and* a PCM temperature

Formulations A and B both store *how much has melted* and nothing else. Ask either of them "how hot is the PCM in segment j?" and the only available answer is T_m — because that is all the state can say. While some solid remains that answer is correct, and it is why the omission went unnoticed for so long.

4.5.2 Where Formulation B breaks, and why it is not a small error

Consider a segment near the well head during charging. The secondary fluid enters at $T_{4c} = 160\,°\mathrm{C}$; the PCM melts at $T_m = 150\,°\mathrm{C}$. That segment sees the hottest fluid, so it melts first. Once its last solid is gone, the resistance network still reports a positive heat flow — the fluid is still 10 K hotter than the melting point, and nothing in Eq. (26) knows the segment is finished.

Formulation B must then do one of two things, and both are wrong:

1. **Keep melting.** A_{melt} grows past the PCM the segment actually contains. This is what the unguarded version did, and it produced local melt fractions up to $\varepsilon_{\mathrm{local}} = 1.78$: the model drew latent heat from material that was not there.
2. **Clip and discard.** Cap A_{melt} at A_{avail} and throw the surplus away. Now the melt fraction is physical, but energy is not conserved — and the fluid leaves the segment at the wrong temperature, because it gave up heat that went nowhere.

The second is what the guarded version did, and the magnitude is the reason this is not a rounding-level concern. At the design point the heat discarded during *discharge* reached 148 % of the heat delivered. Most of what the resistance network asked for was thrown away, and the round-trip figures were computed on what was left.

Notice that no amount of care inside the march can fix this. The defect is in the *state*: a variable that counts melted area has no slot for "liquid above T_m". The physics has to be given somewhere to live.

4.5.3 Why enthalpy, and not temperature

The obvious repair is to carry a PCM temperature alongside the melted area. It is also the wrong repair, and the reason is worth understanding because it recurs throughout phase-change modelling.

At a phase change, temperature is a *bad* state variable: it is not invertible. During melting the PCM sits at T_m while its energy content climbs by the entire latent heat, so T carries no information at all throughout the process that dominates the problem. Carrying (T, A_{melt}) as a pair works, but then the code must decide at every step which of the two is active, keep them consistent, and handle the moments when the active variable changes.

Enthalpy is invertible. Every state of the PCM — subcooled solid, mid-melt, superheated liquid — corresponds to exactly one value of E', and E' moves monotonically as heat is added. So a *single* scalar per segment encodes all three regimes, and the regime is read off from the value rather than tracked separately. Figure 1 is the whole idea in one picture.

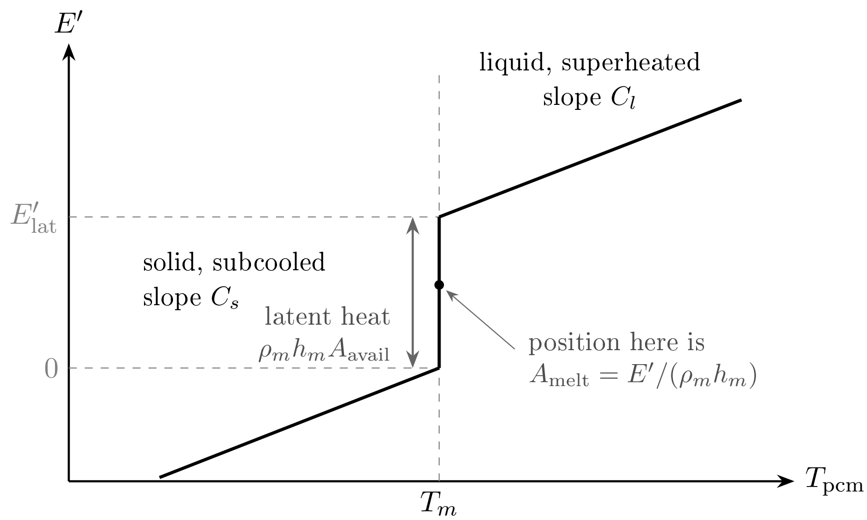

Figure 1: The enthalpy curve, Eq. (43). Reading it as $E'(T)$ it is multivalued at T_m — which is exactly why T cannot be the state. Reading it as $T(E')$, the direction the code uses, it is single-valued everywhere. Position on the vertical segment is the melt fraction.

4.5.4 The state, its capacities, and their sizes

Let E' be the enthalpy per unit tube length of the PCM belonging to one segment, measured from a datum of *fully solid at* T_m. The datum is arbitrary but this choice puts the two branch boundaries at 0 and E'_{lat}, which keeps the tests in the code short.

With $A_{\text{avail}} = V_{\text{well}}/(n_t L_{\text{tube}})$ the PCM cross-section a segment actually owns,

$$E'_{\text{lat}} = \rho_m h_m A_{\text{avail}}, \qquad C_s = \rho_m c_{p,s} A_{\text{avail}}, \qquad C_l = \rho_m c_{p,l} A_{\text{avail}}, \tag{42}$$

are the latent capacity and the two sensible capacities, all per unit tube length. Their numerical values at the design point are worth committing to memory, because they explain the behaviour of the whole formulation:

quantity	value	
A_{avail}	$4.5411\text{e}-3$	m^2
E'_{lat}	2.675	MJ/m
C_s	9.010e3	J/m/K
C_l	1.267e4	J/m/K
C_l/E'_{lat}	$4.74\text{e}-3$	/K

That last ratio is the Stefan number in disguise. One kelvin of superheat is worth 0.47 % of the latent capacity, so the sensible branches are *energetically* minor — with the 10 K approach available here, at most about 4.7 %. A reader could reasonably conclude that the whole formulation is a small correction. It is not, and §4.5.9 explains why: the sensible branches matter through the *driving temperature difference*, not through the energy they store.

The enthalpy curve and the melted area follow directly:

$$T_{\text{pcm}} = \begin{cases} T_m + E'/C_s, & E' < 0 \quad \text{(solid, subcooled)}, \\ T_m, & 0 \le E' \le E'_{\text{lat}} \quad \text{(two-phase)}, \\ T_m + \left(E' - E'_{\text{lat}}\right)/C_l, & E' > E'_{\text{lat}} \quad \text{(liquid, superheated)}, \end{cases} \qquad A_{\text{melt}} = \frac{\min(\max(E', 0),\, E'_{\text{lat}})}{\rho_m h_m}. \tag{43}$$

The clipping in A_{melt} is not a guard bolted on to prevent bad behaviour — it is the definition. Melted area saturates because a segment cannot melt PCM it does not own, and superheat is where the extra energy goes. This is the sense in which $\varepsilon_{\text{local}} \in [0, 1]$ is *structural* in Formulation C: there is no code path that can violate it, so no test is needed.

4.5.5 The segment equation, derived

Equation (52) below is the heart of the march, and it is short enough to look like a definition. It is not: it is the result of writing an energy balance on two different control volumes and then eliminating the quantity they share. This subsection does that in full, because the conductance K that comes out is *not* the $U_i P_i$ a reader would write down unaided, and the difference reaches 33 % at this design point.

The two control volumes. Fix a segment j, spanning $z \in [z_j, z_{j+1}]$ with $\Delta z = z_{j+1} - z_j$. Two control volumes share the tube wall over that interval:

CV 1 — the PCM belonging to segment j: cross-section A_{avail}, length Δz, stationary, closed.

CV 2 — the fluid inside the tube over the same interval: cross-section $A_{\text{flow}} = \pi r_i^2$, open, with mass flow $\dot{m}$ entering at T_j and leaving at T_{j+1}.

Write q'_j for the heat per unit tube length crossing from CV 2 into CV 1. It appears in both balances with opposite sign, and eliminating it is the whole exercise.

Step 1 — the PCM balance. CV 1 is closed and stationary, so the first law reduces to a statement about its enthalpy alone. With E' the enthalpy per unit tube length (§4.5), the enthalpy of the control volume is $E'\Delta z$ and

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(E'_j\,\Delta z\right) \;=\; \underbrace{q'_j\,\Delta z}_{\text{through the tube wall}} \;\; \underbrace{-\left[\dot{Q}_{\mathrm{ax}}\right]_{z_j}^{z_{j+1}}}_{\text{axial conduction, H2}} \;+\; \underbrace{\dot{Q}_{\mathrm{form}}}_{\text{formation, H6}} \;+\; \underbrace{\dot{Q}_{\mathrm{azi}}}_{\text{neighbours, H9}} \;, \tag{44}$$

and the last three terms are dropped by hypothesis. Two of those deletions are worth a number rather than a citation:

- **Axial conduction (H2)** is negligible by an enormous margin. The axial diffusion time over one segment is $\Delta z^2/\alpha_l = 5.4\mathrm{e}9\,\mathrm{s}$ against a $3.6\mathrm{e}4\,\mathrm{s}$ charging window — a ratio of 1.5×10^5. With $\Delta z = 30.5\,\mathrm{m}$ this is not a close call, and it would remain safe at a hundred times the segment count.
- **Azimuthal conduction (H9)** is *not* negligible and is assumed away pending the calculation described in §7.2. At the wellhead the neighbouring cell is $48.9\,\mathrm{K}$ away and a slab estimate gives $\sim 22\,\mathrm{W/m}$ against a useful duty of $\sim 83\,\mathrm{W/m}$. This is the least defensible term in the model.

Dividing by Δz leaves the PCM balance in the form used by the code:

$$\boxed{\frac{\mathrm{d}E'_j}{\mathrm{d}t} \;=\; q'_j} \tag{45}$$

Step 2 — the fluid balance. CV 2 is open. For incompressible single-phase flow (H7) with no viscous dissipation, the energy equation per unit tube length is

$$\underbrace{\rho A_{\mathrm{flow}} c_p\,\frac{\partial T}{\partial t}}_{\text{fluid thermal inertia}} \;+\dot{m}\,c_p\,\frac{\partial T}{\partial z} \;=\; -\,q'(z). \tag{46}$$

The model drops the first term, and this deserves justification rather than assertion. Two comparisons, pointing the same way:

fluid transit time over L_{tube}	2264 s	($u = 1.35\,\mathrm{m/s}$)
charging window t_{ch}	36000 s	
ratio	0.063	
$\rho A_{\mathrm{flow}} c_p\,/\,C_l$	0.257	fluid vs. PCM sensible capacity
$\rho A_{\mathrm{flow}} c_p \Delta T\,/\,E'_{\mathrm{lat}}$	0.0122	fluid swing vs. PCM latent, $\Delta T = 10\,\mathrm{K}$

The fluid re-establishes its profile in one transit, 2264 s, while the PCM state evolves over 10 h: a $16\times$ separation, so quasi-steady is reasonable but *not* asymptotic. The honest reading is that the first 38 min of each half-cycle is a startup transient the model does not resolve, worth about 6 % of the window. What makes this tolerable is the third row: the energy parked in the fluid is only 1.2 % of the latent inventory it is charging, so the transient misplaces very little energy even though it lasts a noticeable fraction of the run.

Setting $\partial T/\partial t = 0$ and using the fact that CV 1 presents a *single* temperature T_{pcm} to the whole segment (H5, the lumped state of §4.5), the driving difference is $T(z) - T_{\text{pcm}}$ and the local flux closes on the network of §3:

$$q'(z) \;=\; U_i\,(2\pi r_i)\,\big[T(z) - T_{\text{pcm}}\big]. \tag{47}$$

Note that $2\pi r_i$ appears, not P_T: U_i was referred to the *inner* area in Eq. (26), and the finned perimeter is already inside it through R'_{outer}. Using P_T here would count the fins twice — which is a compact statement of the v0.1 defect.

Step 3 — integrate across the segment. Substituting Eq. (47) into the steady form of Eq. (46) gives a linear ODE in z with T_{pcm} and U_i held constant over the segment:

$$\dot m c_p \frac{\mathrm{d}T}{\mathrm{d}z} = -2\pi r_i U_i \big(T - T_{\text{pcm}}\big) \quad\Longrightarrow\quad \frac{\mathrm{d}\big(T - T_{\text{pcm}}\big)}{T - T_{\text{pcm}}} = -\frac{2\pi r_i U_i}{\dot m c_p}\,\mathrm{d}z. \tag{48}$$

Integrating from z_j to z_{j+1},

$$T_{j+1} \;=\; T_{\text{pcm}} + \big(T_j - T_{\text{pcm}}\big)e^{-\mathrm{NTU}_j}, \qquad \mathrm{NTU}_j \;=\; \frac{2\pi r_i\,U_i\,\Delta z}{\dot m\, c_p}, \tag{49}$$

which is Eq. (30). It is the single-stream heat-exchanger result: one stream with finite capacity rate $\dot m c_p$, the other an isothermal reservoir of infinite capacity rate, effectiveness $\varepsilon = 1 - e^{-\mathrm{NTU}}$. The reservoir idealisation is exactly what the lumped PCM state licenses.

Step 4 — the closure. The heat the fluid actually gives up over the segment is the enthalpy it loses,

$$q'_j\,\Delta z \;=\; \dot m c_p\big(T_j - T_{j+1}\big) \;\overset{(49)}{=}\; \dot m c_p\big(T_j - T_{\text{pcm}}\big)\big(1 - e^{-\mathrm{NTU}_j}\big), \tag{50}$$

and dividing by Δz gives the conductance that closes Eq. (45):

$$\boxed{q'_j = K_j\big(T_j - T_{\text{pcm}}\big), \qquad K_j = \frac{\dot m\, c_p\big(1 - e^{-\mathrm{NTU}_j}\big)}{\Delta z}.} \tag{51}$$

Combining Eqs. (45) and (51) with the enthalpy curve $T_{\text{pcm}}(E')$ of Eq. (43) closes the system. This is the segment equation:

$$\boxed{\frac{\mathrm{d}E'}{\mathrm{d}t} \;=\; K\left(T_j - T_{\text{pcm}}(E')\right), \qquad K = \frac{\dot m\, c_p\left(1 - e^{-\mathrm{NTU}}\right)}{\Delta z}.} \tag{52}$$

One unknown per segment, E'; everything else is either geometry, a property, or the upstream fluid temperature T_j handed down by the segment before it.

Why K is not U_iP_i. This is the step most easily got wrong, so it is worth isolating. The naive closure takes Eq. (47) with $T(z) \approx T_j$, giving $q'_j \approx 2\pi r_i U_i (T_j - T_{\text{pcm}})$. Comparing:

$$\frac{K_j}{2\pi r_i U_i} \;=\; \frac{1 - e^{-\mathrm{NTU}_j}}{\mathrm{NTU}_j} \;\le\; 1. \tag{53}$$

The two limits say what the difference means:

$\mathrm{NTU} \to 0$: the ratio $\to 1$. The fluid barely cools across the segment, $T(z) \approx T_j$ throughout, and the naive closure is right.

NTU $\to \infty$: $K \to \dot{m}c_p/\Delta z$, a finite limit set by the *fluid*, not the wall. No matter how good the borehole is, a segment cannot absorb more than the stream can carry to it. The naive closure has no such limit and would let T_{j+1} fall below $T_{\rm pcm}$ — heat flowing from cold to hot, at a rate that grows without bound as U_i improves.

At this design point NTU runs from 0.083 to 0.602 over the charge, so Eq. (53) lies between 0.96 and 0.75: **using U_iP_i would overstate the conductance by $4\,\%$ to $33\,\%$**, worst where the melt layer is thinnest and the borehole best. The error is largest exactly where the model is most active, which is why it cannot be waved away as second order.

Two properties inherited from this derivation.

Closure is structural. The code does not evaluate Eq. (51) and Eq. (45) independently. It integrates Eq. (45) over the step and *reports* $q'_j = (E'^{n+1} - E'^{n})/\Delta t$, then drops the fluid temperature by exactly that amount, $T_{j+1} = T_j - q'_j\Delta z/(\dot{m}c_p)$. Whatever the PCM gained, the fluid lost, to machine precision, with no possibility of drift.

K **is frozen over a time step,** $T_{\rm pcm}$ **is not.** K_j is evaluated once per step from the melt thickness at the start of it, whereas $T_{\rm pcm}$ is advanced exactly inside the step by Eq. (54). The consequence is that T_{j+1} equals the ε–NTU result of Eq. (49) only while the segment is on the latent plateau, where $T_{\rm pcm} = T_m$ is genuinely constant. In the sensible branches the two differ at second order in Δt. This is a deliberate choice: it keeps closure exact, which matters more than matching an ε–NTU relation that is itself only quasi-steady.

Two properties of Eq. (52) drive everything that follows:

- It is **nonlinear**, because $T_{\rm pcm}(E')$ is piecewise.
- It is **linear inside each branch**, because each piece of Eq. (43) is affine in E'.

The second property is what makes an exact integration possible, and §4.5.6 takes it up. But first, why an exact integration is not optional.

4.5.6 Why explicit stepping fails — and why nobody noticed until v0.4

Look again at the three branches. On the plateau, $T_{\rm pcm} = T_m$ is *constant*: it does not respond to E' at all. The segment therefore behaves as though it had infinite heat capacity, Eq. (52) reduces to $\mathrm{d}E'/\mathrm{d}t = \text{constant}$, and an explicit Euler step is not merely stable but *exact*.

This is precisely the regime Formulations A and B live in permanently. Explicit stepping was safe in v0.1 and v0.2 for a structural reason, not a lucky one — and that is why the time grid was never examined.

Add the sensible branches and the capacity becomes finite. Equation (52) is now a relaxation equation with time constant C/K, and explicit Euler requires $\Delta t < C/K$. At the design point:

	early ($t = 1\,\mathrm{s}$)	late ($t \to t_{ch}$)
NTU, segment 1	0.592	0.083
K [W/m/K]	64.3	11.5
C_l/K [s]	197	1106
time-grid step Δt [s]	0.31	8491
$\Delta t\,K/C_l$	0.002	7.7

The grid is logarithmic, so it is fine at the start and violates the limit by almost an order of magnitude at the end. The first implementation duly diverged: the oscillation fed back through the fluid temperature and drove the secondary fluid to 261 K, some 170 K below anything physical.

This is a good failure to have seen once. It was loud, it was immediate, and it was diagnosable. The dangerous version of the same error is a step ratio of 1.5 rather than 7.7: still unstable in principle, but slowly enough to look like a plausible answer.

4.5.7 Branchwise-exact integration

Rather than shrink Δt — which would multiply the cost of every run by $\sim 10^3$ — exploit the linearity within each branch and integrate in closed form. Holding the fluid temperature T_j fixed across the step:

$$\underbrace{E'(t+\Delta t) = E' + K\,(T_j - T_m)\,\Delta t}_{\text{plateau: linear}}, \qquad \underbrace{E'(t+\Delta t) = E'_{\text{eq}} + \big(E' - E'_{\text{eq}}\big)e^{-K\Delta t/C}}_{\text{sensible: exponential relaxation}}, \tag{54}$$

with the equilibrium enthalpy and the relevant capacity

$$\big(C,\ E'_{\text{eq}}\big) = \begin{cases} \big(C_s,\ C_s\,(T_j - T_m)\big), & E' < 0, \\ \big(C_l,\ E'_{\text{lat}} + C_l\,(T_j - T_m)\big), & E' > E'_{\text{lat}}. \end{cases} \tag{55}$$

E'_{eq} is the enthalpy at which the PCM would sit in equilibrium with the fluid — the state the segment relaxes toward and never quite reaches. The exponential is unconditionally stable and monotone: no step size, however large, can overshoot it. That is the property explicit Euler lacks.

Crossing a branch boundary. A step may start in one branch and end in another. The step is then split at the crossing. Let E'_b be the boundary being approached (0 or E'_{lat}). The crossing time is

$$t^* = \begin{cases} \dfrac{E'_b - E'}{K\,(T_j - T_m)}, & \text{from the plateau,} \\[2ex] \dfrac{C}{K}\,\ln\!\left(\dfrac{E' - E'_{\text{eq}}}{E'_b - E'_{\text{eq}}}\right), & \text{from a sensible branch,} \end{cases} \tag{56}$$

and if $t^* < \Delta t$ the state is advanced to E'_b, the remaining $\Delta t - t^*$ is integrated in the adjacent branch, and the test is repeated. At most a handful of crossings can occur in one step, so the loop is bounded.

The heat reported to the fluid is then

$$q'_j = \frac{E'^{\,n+1} - E'^{\,n}}{\Delta t}, \tag{57}$$

by definition, not by a separate flux evaluation. Whatever the segment gained, the fluid lost. Closure is exact to machine precision and cannot drift.

A bug worth studying: the sticky boundary

The branch test in the code is written with closed intervals, $0 \le E' \le E'_{\text{lat}}$, so a state landing *exactly* on E'_{lat} is still classified as two-phase. Follow the logic: the plateau branch computes $t^* = (E'_{\text{lat}} - E')/\text{rate} = 0$, finds $t^* < \Delta t$, advances the state to E'_{lat} — where it already is — and subtracts zero from the remaining time. The next iteration does the same. The segment is

pinned at the boundary for ever and never enters the liquid branch.
The symptom was not a crash. It was every segment reporting a melt fraction of exactly 1.000 and a superheat of exactly zero — results that look *excellent*. The tell was a single diagnostic line printing "sensible energy stored: 0.000", a quantity that had no business being identically zero.
The fix is to step just past the boundary rather than onto it,

$$E' \leftarrow E'_{\text{lat}} + \epsilon, \qquad \epsilon = 10^{-9} \max\bigl(E'_{\text{lat}}, 1\bigr)\,,$$

so the next iteration selects the liquid branch. The general lesson: when a solver switches on the sign of a quantity, decide deliberately which side of the boundary the equality belongs to, and make sure the state cannot come to rest on it.

Verification. The scheme was checked against a brute-force explicit integration of Eq. (52) using 200000 substeps, over a charge and a discharge, including steps that cross branch boundaries. Agreement is 3×10^{-13} relative — machine precision. This is a genuine verification: the two calculations share no code path beyond Eq. (43).

4.5.8 Recovering the melt-layer thickness

The resistance network needs δ, not A_{melt}. Inverting Eq. (39) — which excludes the fin metal from the melted PCM — gives δ as described in §4.4, and the network of §3 is otherwise unchanged in form.

One thing *is* changed: the driving temperature. In Formulations A and B the outer boundary of the resistance chain sits at T_m. In Formulation C it sits at T_{pcm}, which equals T_m only while some solid remains. This single substitution is the mechanism of the next subsection.

4.5.9 Sensible heat is self-levelling

Here is why a 4.7 % energy term produces a first-order change in the answer.

Consider again the well-head segment that melts first. Under Formulation B it finished melting and then continued to absorb heat at the full driving difference $T_j - T_m$, because its boundary temperature was pinned at T_m regardless of state. It monopolised the fluid's heat. The segments downstream, which had not finished, received whatever was left.

Under Formulation C the same segment starts to superheat the instant its solid is gone. T_{pcm} rises, $T_j - T_{\text{pcm}}$ collapses, its demand falls away, and the heat continues down the well to segments that still have solid to melt. The mechanism is a negative feedback, and it is automatic: nothing in the code distributes the heat, and no parameter was tuned.

The effect on the distribution of melt is large:

formulation	$\varepsilon_{\text{local}}$ at ten stations										spread
B (latent only)	1.78	0.73	0.69	0.71	0.77	0.85	0.94	1.03	1.12	1.23	1.321
C (enthalpy)	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	**0.078**

where the spread is $\max_j \varepsilon_{\text{local}} - \min_j \varepsilon_{\text{local}}$ over the 100 segments at end of charge. Two separate things improved. The values above 1 are gone, because Eq. (43) makes them impossible. The values well below 1 are gone too, and that is the physical result: PCM that never melts is *dead volume*

— it occupies borehole, costs money, and carries no energy through the round trip. Formulation C says there is far less of it than Formulation B implied.

4.5.10 What Formulation C still does not do

Three limitations, stated plainly because they bound how far the result can be pushed.

One temperature per segment. E' is a lumped quantity: all the PCM in a segment is at a single T_{pcm}. There is no radial resolution *within* a phase, so superheat appears instantaneously and uniformly across the whole liquid annulus rather than diffusing outward from the tube. The model therefore reaches its superheated state sooner than the real material would, and the self-levelling above is an upper bound on the effect.

H1 is inherited, not repaired. The quasi-steady melt layer of §4.2 is still assumed: the conduction profile through the liquid is taken as instantaneously steady. This is justified by $\mathrm{Ste} = 0.047$ and is a good approximation here, but it is an approximation, and it is the same one in all three formulations.

H3 is now *saturated*. Because $\varepsilon_{\mathrm{local}} \to 1$ over essentially the whole well, $\delta \to \delta_{\mathrm{merge}}$ everywhere — the melt fronts of neighbouring legs are touching. The annular conduction resistance is being evaluated at the exact limit of its validity. §7.1 takes this up, because it now carries a design decision.

Formulation C is nonetheless the first of the three that is *closed*: every joule the resistance network delivers has a place to go, and the melt fraction cannot leave $[0, 1]$. Whatever is wrong with the answer is now wrong in the heat transfer, not in the bookkeeping.

Source: `pcm_capacities`, `pcm_state`, `advance_segment`, `march_h`.

4.6 Latent inventory and the melted fraction

The melted volume per tube and the melted fraction of the borehole inventory are

$$V_{\mathrm{melt}} = \sum_{j=1}^{n_s} A_{\mathrm{melt}}^{(j)} \, \Delta z, \qquad \varepsilon_{\mathrm{PCM}} = \frac{V_{\mathrm{melt}} \, n_t}{V_{\mathrm{well}}}, \tag{58}$$

with V_{well} from Eq. (4) and the multiplier n_t, not $2n_t$, for the reason given after Eq. (2).

The density ρ_m in Eqs. (36)–(38) is the *same in both directions*, and is taken as the solid density ρ_s. The thermal conductivity k_m by contrast *is* directional, k_l while melting and k_s while freezing, because the layer adjacent to the tube is liquid in the first case and solid in the second.

The asymmetry is deliberate. Making the latent inventory directional as well would bank energy at ρ_l and return it at ρ_s, yielding $\rho_s/\rho_l = 1.069$ times more energy out than in — a 6.9 % round-trip gain from bookkeeping alone. Carrying the front as a *solid-equivalent* area instead makes $\varepsilon_{\mathrm{PCM}}$ a true melted *mass* fraction and makes stored energy equal to available energy. The price is that the 6.9 % volume change on melting is not represented: the liquid layer is about 2.3 % thicker in radius than Eq. (39) reports, so Eq. (20) mildly under-predicts the melt-layer resistance.

Source: `thums_core/stefan.py`; `config.PCM.rho_latent`.

5 Hydraulics, well-field sizing and performance indices

5.1 Pressure drop and pumping power

Properties are evaluated at the mean of the inlet and outlet temperatures. With the flow area $A_{\text{flow}} = \pi r_i^2$,

$$u = \frac{\dot{m}}{\rho A_{\text{flow}}}, \qquad \text{Re} = \frac{\rho\, u\, D_i}{\mu}, \tag{59}$$

and the Darcy friction factor is taken as laminar or Blasius,

$$f = \begin{cases} \dfrac{64}{\text{Re}}, & \text{Re} < 2000, \\[2mm] \dfrac{0.3164}{\text{Re}^{1/4}}, & \text{Re} \le 10^5, \end{cases} \tag{60}$$

giving the Darcy–Weisbach frictional drop

$$\Delta p_{\text{fric}} = f\, \frac{L_{\text{tube}}}{D_i}\, \frac{\rho u^2}{2}. \tag{61}$$

Minor losses are added with a lumped coefficient for one hairpin — a sharp-edged entrance, two 180° return bends and an exit:

$$K_{\text{tot}} = K_{\text{ent}} + 2K_{\text{bend}} + K_{\text{exit}} = 0.5 + 2(2.2) + 1.0 = 5.9, \qquad \Delta p_{\text{min}} = K_{\text{tot}}\, \frac{\rho u^2}{2}. \tag{62}$$

The total drop and the hydraulic power for one tube are then

$$\Delta p = \Delta p_{\text{fric}} + \Delta p_{\text{min}}, \qquad \dot{W}_{\text{pump,tube}} = \frac{\dot{m}\, \Delta p}{\rho}. \tag{63}$$

The three coefficients in Eq. (62) are hard-coded literals rather than configuration parameters, and the source describes them as example values. They are not derived from the completion geometry and cannot be varied from `Case`. At the design point Δp_{min} is small beside Δp_{fric} because $L_{\text{tube}}/D_i \approx 10^5$, but the asymmetry is worth knowing: any sensitivity study on pumping power is varying only the friction term. The field total scales by the tube and well counts,

$$\dot{W}_{\text{pump}} = \dot{W}_{\text{pump,tube}}\, n_t\, N_{\text{wells}}. \tag{64}$$

Two properties of Eq. (60) deserve note. The smooth-tube assumption sets the roughness to zero, which is optimistic for a well completion. And no correlation is supplied above $\text{Re} = 10^5$; the Blasius branch is simply extrapolated, which is where discharge cases with elevated flow will land.

Source: `_legacy_physics.calculate_pressure_drop`.

5.2 Well-field sizing

Two questions, not one

A well field must satisfy *two* independent requirements, and the model names one number for each. They are easy to confuse, so state them as questions before writing any algebra:

symbol	the question it answers	how it is found
N_{rate}	Can the field **absorb** ΔE *within* t_{ch}?	root solve on the march
N_{capacity}	Can the field **contain** ΔE *at all*?	closed form
N_{wells}	**Both must hold.**	max of the two

The distinction is between a *rate* and a *stock*. A field could have ample PCM but transfer heat into it too slowly to finish within the charging window; or it could transfer heat beautifully into a store too small to hold the energy. Neither failure implies the other, so neither criterion implies the other, and the design must clear both.

The rate criterion. This is the only requirement v0.1 imposed. Writing $\Delta E_{\mathrm{del}}(N_{\mathrm{wells}})$ for the energy the field takes up over t_{ch},

$$\Phi_{\mathrm{rate}}(N_{\mathrm{wells}}) \;\equiv\; \frac{\Delta E_{\mathrm{out,HP}}}{\Delta E_{\mathrm{del}}(N_{\mathrm{wells}})} - 1 \;=\; 0. \tag{65}$$

There is no closed form for ΔE_{del}: it comes from marching the whole well, so N_{rate} carries *everything thermal* — fin count and geometry, wall conductivity, melt-layer resistance, NTU, the cascade, and the length of the charging window. Every root evaluation is a full march, which is why this criterion is expensive.

The capacity criterion. The requirement v0.1 omitted. It asks only whether the PCM in the ground can hold the energy, with no reference to how fast it arrives:

$$\boxed{N_{\mathrm{wells}} \;=\; \max\bigl(N_{\mathrm{rate}},\; N_{\mathrm{capacity}}\bigr), \qquad N_{\mathrm{capacity}} = \frac{\Delta E_{\mathrm{out,HP}}}{E_{\mathrm{well}}},} \tag{66}$$

with the capacity of one well from latent *and* sensible heat:

$$E_{\mathrm{well}} = \rho_m V_{\mathrm{well}} \left[h_m + c_{p,l}\,(T_{3c} - T_m)\right]. \tag{67}$$

The sensible term is the zero-resistance limit: with no thermal resistance the liquid would reach the charging inlet temperature. Note what is *not* in Eq. (67) — no U_i, no δ, no NTU, no t_{ch}. It is a statement about volume and material properties and nothing else, which makes it a **hard lower bound**: no design, however good thermally, and no charging window, however long, can do better. It is also closed form, so unlike the constraint it replaces it costs nothing to evaluate.

At the design point $N_{\mathrm{rate}} = 11.255$ and $N_{\mathrm{capacity}} = 11.573$, so capacity binds and $N_{\mathrm{wells}} = 11.573$. With the legacy wall conductivity the rate criterion dominated instead, which is why the omission stayed invisible for so long.

A reporting rule: N_{capacity} is a bound, not a result

Equation (66) is a *maximum*, and that governs how its parts may be read. N_{capacity} is the seductive one — closed form, smooth, responsive to every geometric parameter — and it is wrong to read on its own. Whenever the rate criterion binds, N_{capacity} moves while the answer does not, and it can move the opposite way.

The failure is not hypothetical. §7.1 records a fin sweep in which N_{capacity} falls monotonically while N_{wells} rises by 42 %. Two properties of Eq. (67) explain it: $E_{\mathrm{well}} \propto V_{\mathrm{well}}$, and V_{well} counts whatever the metal does not occupy — so $N_{\text{capacity}} \cdot V_{\text{well}}$ is invariant under any change to the fins, and the column is pure volume bookkeeping.

The model therefore reports through `sizing_report`, which leads with N_{wells}, names the criterion that set it, gives the margin by which the other is slack, and labels N_{capacity} as a lower bound. At the design point the margin is 2.7 %: the design sits *on* the crossover, so any parameter sweep switches criterion partway through, and the report warns when this is so.

Two names that have been retired

Earlier versions exposed *four* names for these two numbers, and two of them were traps. Both now raise an error carrying this explanation rather than returning a number:

`N_inventory` was a silent *alias* for N_{capacity} in v0.4 — the two compared equal and could not be told apart. Worse, in **v0.3 the same name meant something else**: the well count at which $\varepsilon_{\text{PCM}}(N) = 1$, a marched quantity requiring its own root solve. That condition is not a capacity statement at all but "the well count at which an unthrottled 10 h charge exactly saturates the store", and it allowed the field to over-charge by 7 % relative to the requirement. *A v0.3 printout of `N_inventory` is therefore not comparable with a v0.4 one*, and nothing in the output said so.

`N_heat` was the dictionary key for the rate criterion while the printed report called it "charge-rate" and the sweep column headed it `N_rate`. Three labels for one quantity, and the obvious guess `N_rate` raised a bare `KeyError`. The rate criterion is now `N_rate` everywhere.

The surviving vocabulary is three words: **rate**, **capacity**, **the maximum of the two**.

Why the field is not sized on the discharge

It is natural to ask whether N_{wells} should instead be converged on the discharge — the duty that must actually be met — with the flow converged on the charge. It cannot, and the reason is structural: *sizing must act on whichever variable is free.*

During charging the flow is pinned by the specified glide, $\dot{m}_w^{\text{ch}} = \dot{Q}_{\text{out,HP}}/(c_{p,w}\Delta T_{3C,2C})$, so N_{wells} is the only freedom and the charge can size it. During discharging N_{wells} is already fixed, so the flow is the freedom and the discharge sizes that instead.

Pinning the discharge flow to its own glide as well — $\Delta T_{2D,3D}$ is tied to $\Delta T_{3C,2C}$ by Eq. (10) — over-determines the problem, and measurably so:

N_{wells}	ε_{PCM}	$\Delta E_{\text{dc}}/\Delta E_{\text{in,ORC}}$
8	1.505	0.9285
12.74	0.999	0.9776
18	0.719	0.9925
24	0.541	0.9960
40	0.325	0.9962

The delivered energy saturates at 0.996 and never reaches unity for *any* well count, because the fluid leaves slightly short of the nominal outlet temperature and so carries about 0.4 % less energy per unit mass than $c_p\Delta T$ presumes. There is no root. Letting the discharge flow float by the required 1.3 % is precisely what the flow loop exists to do, and the v0.1 arrangement is therefore correct and is retained.

Comparison with the v0.1 ideal well count

The v0.1 notebook printed `Ideal number of wells:  12.215` from

$$m_{\text{ideal}} = \frac{\Delta E_{\text{out,HP}}}{h_m + c_{p,l}\,(T_{3c} - T_m)}, \qquad m_{\text{well}} = V_{\text{well}}\,\rho_l, \tag{68}$$

which is Eq. (67) with ρ_l in place of ρ_m. It carries no thermal resistance and no losses, so it is a genuine **lower bound**: the best any design could achieve. Designs with high ε_{PCM} are desirable precisely because they approach it — the PCM in the ground is being used rather than merely occupying the borehole.

Correction to version 0.3 of this document. v0.3 stated that $N_{\text{inventory}}/N_{\text{ideal}} = 1.0428$ agreed with $1 + \text{Ste} = 1.0474$, and attributed the gap to the sensible heat the model then omitted. That comparison was made on *inconsistent bases* and the agreement was a coincidence. Equation (68) uses ρ_l and credits sensible heat; the v0.3 model used ρ_s and credited none. On a common basis:

variant	N
Eq. (68) as coded: $h_m + c_{p,l}\Delta T$, ρ_l	12.371
latent only, ρ_s	12.121
$h_m + c_{p,l}\Delta T$, ρ_s — **consistent with v0.4**	**11.573**

The two offsetting differences were a factor 0.955 from the sensible credit and 0.936 from the density, net 0.893. With v0.4 the question disappears: N_{capacity} *is* Eq. (68) evaluated on the model's own conventions, so the comparison is an identity rather than a check.

Source: `thums_core/sizing.py`; reported by `sizing_report`.

Discharge flow

The discharge flow is then found so that the field delivers the ORC's heat demand over t_{dc}. With $\mathcal{R} = \dot m_1^{\text{dc}}/\dot m_1^{\text{ch}}$ the ratio of per-tube flows,

$$\frac{\Delta E_{\text{in,ORC}}}{\Delta E_{\text{dc}}(\mathcal{R})} - 1 \;=\; 0, \tag{69}$$

solved by bisection on $\mathcal{R}$. Under Formulations B and C the discharge march starts from the melt distribution charging left behind, so Eq. (69) is bounded by the stored inventory; under Formulation A it starts from a bare tube in fully solid PCM and is not.

5.3 Performance indices

The round-trip efficiency, with and without parasitics:

$$\eta_{\text{RTE}} = \frac{\left(\dot W_{\text{el,out}} - \dot W_{\text{pump}}^{\text{dc}}\right) t_{\text{dc}}}{\left(\dot W_{\text{el,in}} + \dot W_{\text{pump}}^{\text{ch}}\right) t_{\text{ch}}}, \tag{70}$$

$$\eta_{\text{RTE}}^0 = \frac{\dot W_{\text{el,out}}\, t_{\text{dc}}}{\dot W_{\text{el,in}}\, t_{\text{ch}}}. \tag{71}$$

By Eq. (12), η_{RTE}^0 depends only on the two cycle efficiencies, the four electrical and mechanical efficiencies, and λ. It is therefore invariant to every storage-model change in this document, and a useful check that a modification has not leaked into the cycles.

The remaining indices are

$$\varepsilon_{\text{PCM}} = \frac{V_{\text{melt}} n_t}{V_{\text{well}}} \qquad \text{melted fraction of inventory,} \tag{72}$$

$$E_{\text{well}} = \frac{\Delta E_{\text{in,ORC}}}{N_{\text{wells}}} \qquad \text{energy delivered per well,} \tag{73}$$

$$\rho_E = \frac{E_{\text{well}}}{V_{\text{well}}} \qquad \text{volumetric energy density,} \tag{74}$$

$$f_{\text{pump}} = \frac{\dot{W}^{\text{ch}}_{\text{pump}} + \dot{W}^{\text{dc}}_{\text{pump}}}{\dot{W}_{\text{el,out}}} \qquad \text{parasitic fraction,} \tag{75}$$

$$\eta_{\text{storage}} = \frac{\Delta E_{\text{discharged}}}{\Delta E_{\text{stored}}} \qquad \text{recovered fraction, Formulations B and C.} \tag{76}$$

Source: `thums_core/kpi.py`; assembled in `system.run_cycle`.

A symbol that carries two meanings. E_{well} above is formed from $\Delta E_{\text{in,ORC}}$, the energy the ORC *consumes*: a duty per well. The E_{well} of Eq. (67) is the energy one well can *hold*: a capacity. The code keeps them in different keys (`kpis['E_well']` and `detail['E_well_kJ']`) but under one name, and they are not equal — 4.518 MWh against 4.744 MWh at the design point.

The gap is not a rounding artefact and not a property of ε_{PCM}. Whenever the capacity criterion binds, $N_{\text{wells}} = \Delta E_{\text{out,HP}}/E^{\text{cap}}_{\text{well}}$, so

$$\frac{E^{\text{duty}}_{\text{well}}}{E^{\text{cap}}_{\text{well}}} = \frac{\Delta E_{\text{in,ORC}}}{\Delta E_{\text{out,HP}}} = \frac{1}{1+\lambda} = 0.9524, \tag{77}$$

exactly the assumed storage-loss surplus of §2.3, and the observed ratio is 0.9524. The two symbols are therefore related by a *model input*, which makes the shared name actively misleading. Renaming one of them is on the list.

6 Cyclic operation

Everything in §5 is a *first-cycle* result. The store begins at $E' = 0$ — fully solid at exactly T_m, with no subcooling — and the cycle does not return there. A plant repeating a daily schedule never sees that state again after the first day, so the well counts reported above are cold-start numbers.

The intended duty is 10 h **charge,** 2 h **rest,** 10 h **discharge,** 2 h **rest**.

6.1 The idle periods are exact no-ops

With no flow, NTU $\to \infty$ but the conductance of Eq. (51) carries $\dot{m}$ as a prefactor:

$$K = \frac{\dot{m} c_p \left(1 - e^{-\text{NTU}}\right)}{\Delta z} \xrightarrow[\dot{m}\to 0]{} 0, \tag{78}$$

so $\mathrm{d}E'/\mathrm{d}t = K(T_j - T_{\text{pcm}}) = 0$ identically. Hypotheses H2 (no axial conduction), H6 (adiabatic wall) and H9 (no azimuthal exchange) leave no other path by which a segment can exchange heat with anything. The 2 h rests therefore change nothing: the 10/2/10/2 schedule is arithmetically identical to 10/10 in this model.

That is worth reading as a diagnostic rather than a convenience. Physically, a 2 h rest lets superheated liquid near the tube conduct outward into colder PCM and lets the front relax; the lumped state of H5 cannot represent that either. The rests doing nothing is a statement about the strength of four assumptions, not about the store.

6.2 Why sizing cannot simply be moved to cyclic steady state

At cyclic steady state (CSS) the state returns to itself, so the enthalpy change over a complete cycle is zero. The model has no loss path, and therefore

$$\boxed{\eta_{\text{storage}}(\text{CSS}) \;\equiv\; 1} \tag{79}$$

exactly — for every well count and every flow. Computed values agree to eight decimal places across N_{wells} from 11 to 20 and flow ratios from 0.92 to 1.03. Equation (79) is not a result about the store; it is energy conservation applied to a closed cycle in an adiabatic model, and it is worth stating only because of the two consequences that follow.

Consequence 1: the loss surplus is unattainable

The budget of Eq. (12) requires $\Delta E_{\text{out,HP}} = (1+\lambda)\Delta E_{\text{in,ORC}}$. With Eq. (79) the ratio of CSS charge to requirement pins at

$$\frac{\Delta E^{\text{CSS}}_{\text{charge}}}{\Delta E_{\text{out,HP}}} \;=\; \frac{1}{1+\lambda} \;=\; 0.95238 \tag{80}$$

for *every* N_{wells}. The sweep returns 0.95247 at $N_{\text{wells}} = 12$ and 0.95231 at $N_{\text{wells}} = 20$: flat. There is no root, and the CSS sizing problem as posed in §5.2 has no solution at any well count.

It is worth being precise about where the surplus goes on the first cycle, since nothing in the model loses it. λ enters in exactly one place — the line $\Delta E_{\text{out,HP}} = (1+\lambda)\Delta E_{\text{in,ORC}}$ — and propagates only to the charge target, the charging mass flow, and $\dot{W}_{\text{el,in}}$. Nothing checks that the surplus is lost. It is a requirement to *deposit* more, and on the first cycle the extra is simply **parked in the store** as residual melt:

	kJ, field total
charged, actually delivered	1.97546e8
discharged	1.88298e8
difference	9.24816e6
$\lambda\,\Delta E_{\text{in,ORC}}$	9.41140e6
ratio	0.983

The first cycle gets away with it only because it starts from an empty store. At CSS there is nowhere left to park it.

A correction to §5.2 as written at v0.4b. That section reported $\eta_{\text{storage}} = 0.9532$ against the assumed $1/(1+\lambda) = 0.9524$ and offered the agreement as independent corroboration of the capacity correction. It is not independent. The discharge loop solves so that the delivered energy equals $\Delta E_{\text{in,ORC}}$, and the charge is sized so the absorbed energy equals $(1+\lambda)\Delta E_{\text{in,ORC}}$; therefore $\eta_{\text{storage}} \equiv 1/(1+\lambda)$ identically whenever both loops converge. It is a tautology and should not have been offered as evidence.

Consequence 2: with $\lambda = 0$ the rate criterion goes degenerate

Setting $\lambda = 0$ removes the contradiction — the target becomes attainable — but it does not restore the ability to size. By Eq. (79) the CSS charge and the CSS discharge are then the *same number*, so the two sizing equations collapse into one. It fixes the *flow ratio* and says nothing whatever about N_{wells}:

N_{wells}	flow ratio	charge/req	discharge/req
11.0	1.0740	1.00000	1.00000
14.0	1.0280	0.99994	0.99994
20.0	0.9965	0.99994	0.99994

This is the same structural failure that §5.2 records for the discharge side, now appearing on the charge side as well. Cycling is what exposes it. Note also that adding a loss mechanism — relaxing H6 — would remove the *contradiction* of Consequence 1 but not this degeneracy: at CSS the charge would then exceed the discharge by exactly the loss, and the one remaining equation would still pin the flow ratio rather than N_{wells}.

6.3 The reformulation: specify, simulate, report

The resolution is to stop asking the energy balance to determine N_{wells}. Specify the hardware, march to CSS, and report what comes out:

quantity	how it is set
N_{wells}	$\Delta E_{\text{out,HP}}/(\rho_m V_{\text{well}} h_m)$, latent heat alone
$\dot{m}^{\text{ch}}$	pinned by the charging glide
$\dot{m}^{\text{dc}}$	pinned by the discharging glide

Nothing is solved. There is no root-find anywhere in `simulate_css`, so the question of convergence does not arise; what was an unsatisfiable constraint becomes a reported output — the deviation of the delivered energy, and hence of the net electrical output, from its target. Sizing on latent heat alone *overestimates* N_{wells}, because it ignores the sensible capacity the store also has; that is deliberate and conservative.

This framing also dissolves an older objection. §5.2 argues that the discharge flow cannot be pinned to its glide, because the delivered energy then saturates below the requirement for any well count and leaves no root. Under the reformulation that saturation is not a failure: it *is* the answer.

Result at the design point

N_{wells}	11.5441	latent heat alone, not solved
$\dot{m}^{\text{ch}}$, $\dot{m}^{\text{dc}}$	0.9655, 0.9698	kg/s per leg-pair
flow ratio	1.0044	a consequence, not a solve
cycles to CSS	28	
η_{storage}	1.000000	Eq. (79)
required $\Delta E_{\text{in,ORC}}$	1.88228e8	kJ
delivered at CSS	1.77417e8	kJ
deviation	**-5.736**	%
implied net output	0.9426	MWe against a target of 1.0000

The shortfall is a rate limit, not an inventory limit

At CSS the store melts *completely* — $\varepsilon_{\text{local}} = 1.000$ at every segment at end of charge — but returns only to a mean of 0.154. It never fully refreezes within the 10 h discharge window, and that residual is what limits delivery. The design curve makes the diagnosis explicit:

N_{wells}	delivered/req	MWe	ε end charge	ε end discharge
11.544	0.9426	0.9426	1.000	0.154
12.500	0.9540	0.9540	1.000	0.209
14.000	0.9662	0.9662	1.000	0.288
16.000	0.9777	0.9777	1.000	0.376
20.000	0.9931	0.9931	1.000	0.505
26.000	1.0082	1.0082	1.000	0.629

The residual melt fraction *rises* with N_{wells}. That is the signature of a rate-limited discharge: more wells means each is worked less hard, so proportionally less of each one refreezes. Were the limit an inventory limit, the residual would fall. The lever is therefore on the discharge side — flow, window, or conductance — and not more PCM. The target is met near $N_{\text{wells}} \approx 25$, about $2.2\times$ the latent-only count.

How much of this to believe

The $-5.7\,\%$ deviation is a model result and inherits every assumption of §7. H9 is worth of order $25\,\%$ of the useful duty at the wellhead (§7.2); H3 is saturated over $99\,\%$ of the well (§7.1); the time grid carries a one-sided $-0.3\,\%$ bias. A $6\,\%$ deviation sits inside that band. It should be reported as *"the store is discharge-rate limited by roughly* $6\,\%$ *at this sizing"* and not as a calibrated shortfall.

6.4 A defect that cycling exposed

The discharge marches from the opposite end of the hairpin, because the flow reverses between half-cycles. The melting-temperature cascade is therefore mirrored with respect to the march index: $T_m^{\text{dc}}(i) = T_m^{\text{ch}}(n_s - 1 - i)$. But E' is measured from a datum of *solid at the local* T_m (§4.5), and until v0.5 the state array was handed to the discharge **without** being mirrored. The datum therefore moved by up to 48.9 K across the half-cycle boundary.

The symptom was invisible until T_{pcm} was plotted rather than $\varepsilon_{\text{local}}$: at the start of the discharge the model reported PCM at 176.6 °C against a charging inlet of 160.0 °C — hotter than any fluid that had ever touched it. Read against the charge cascade the same state tops out at exactly 160.00 °C, which is correct, and that contrast is what identified the error.

Two changes fix it. The discharge cascade is now built as the exact reverse of the charge cascade, which additionally removes a one-layer (6.111 K) mismatch that `layer_map` introduces whenever n_s is not a multiple of N_{lay}; and the state array is mirrored at each half-cycle boundary.

	before	after	change
N_{wells} (first cycle)	11.2551	11.2551	—
flow ratio	1.0262	1.0473	$+2.06\,\%$
η_{RTE}	0.41475	0.41391	$-0.20\,\%$
η_{storage}	0.95318	0.95266	$-0.055\,\%$
CSS deviation	$-3.34\,\%$	$-5.74\,\%$	

N_{wells} is unchanged because the field is sized on the charge, which was never affected; everything downstream of the discharge moves. The effect on the CSS result is much larger than on the single cycle, which is what one would expect of an error that compounds across repeated half-cycle boundaries.

7 Violated assumptions, duplicated definitions, absent terms

This section is part of the model specification, not an appendix to it. Every item below is either an assumption the operating regime violates, a physical quantity the code defines twice, or a term omitted altogether. A reader reproducing these results needs all three.

7.1 H3 is saturated, and it now carries a design decision

The limit was being tested at the wrong radius

Hypothesis H3 takes the melt region to be a concentric annulus whose front never meets the front of a neighbouring leg. Until this version the geometric check guarding it compared $r_e + \delta$ against the *borehole wall*. That is the wrong radius. The $2n_t = 4$ legs share the borehole cross-section equally, so each owns

$$\pi r_{\text{cell}}^2 = \frac{\pi R^2}{2n_t}, \qquad \text{i.e.} \qquad r_{\text{cell}} = \frac{R}{\sqrt{2n_t}}, \qquad \delta_{\text{merge}} = r_{\text{cell}} - r_e, \tag{81}$$

with $R = D_{\text{well}}/2$. Numerically $r_{\text{cell}} = 44.45\,\text{mm}$ against a wall at $88.90\,\text{mm}$, so the check permitted $\delta = 67.82\,\text{mm}$ where the true limit is $23.37\,\text{mm}$ — a factor 2.9 too permissive, and therefore silent at every design point ever run. Equation (81) is now `Case.r_cell` and `Case.delta_merge`, and the plots, the diagnostics and the sizing report all read the same limit.

Under Formulation C the limit cannot be exceeded — it is sat on

Correcting the radius turns out not to produce a violation, for a reason worth stating precisely:

$$A(\delta_{\text{merge}}) \equiv A_{\text{avail}}, \tag{82}$$

identically — a leg owns exactly the PCM inside its own cell, so melting all of it puts the front exactly on the cell boundary. Since Formulation C caps A_{melt} at A_{avail} (§4.5), it follows that $\delta \le \delta_{\text{merge}}$ always. $\varepsilon_{\text{local}} = 1$ *and "the fronts merge" are the same line.*

What the corrected diagnostic reveals instead is that the design *sits* on that line: at the v0.4 design point $\delta/\delta_{\text{merge}} = 1.000$ over 99 % of the well. The annular conduction resistance is being evaluated at the exact limit of its validity almost everywhere. Two errors follow, and both point the same way:

1. The resistance $\ln(1 + \delta/r_e)\,/2\pi k_m$ assumes the full azimuth $2\pi\,(r_e + \delta)$ conducts. Past merge it does not — a neighbour has taken part of it — and the residual solid sits in the *corners* between legs, on a longer and narrower path. Late-stage resistance is **understated**.
2. Fins enter U_i only as added *surface at the tube*, through η_f and P_T (§3.3). They are not represented as radial conduction paths reaching into that corner PCM — which is the single thing a fin does best in this regime. The benefit is **uncredited**.

Why this now matters: the fin count

This ceased to be a bookkeeping concern when it began to decide geometry. Sweeping the fins gives:

configuration	N_{capacity}	N_{rate}	N_{wells}	binding
24×7.50 mm	11.573	11.263	**11.573**	capacity
16×7.50 mm	11.348	11.287	**11.348**	capacity
12×7.50 mm	11.239	11.529	**11.529**	rate
8×7.50 mm	11.132	12.208	**12.208**	rate
bare tube	10.924	16.569	**16.569**	rate

Read the N_{capacity} column alone and the conclusion is that fins make matters worse — it falls monotonically as fin metal is removed. That column contains *no heat transfer at all*: it is $\Delta E / (\rho V_{\text{well}} [h_m + c_{p,l} \Delta T])$, and removing a fin returns its metal volume to the PCM. Indeed $N_{\text{capacity}} \cdot V_{\text{well}}$ is constant to machine precision across the whole sweep. The real answer, N_{wells} = max of the two criteria, moves the other way: removing the fins costs 42 % more wells. §5.2 states the reporting rule that follows.

The optimum sits where the two criteria cross, at 16×7.5 mm — so the design point is mildly over-finned, by about 2 % in well count. But given the two biases above, that figure should be read as a **floor** on the fin count rather than an optimum. Settling it requires a two-dimensional conduction calculation in the cell, which is the most consequential piece of open work in this model.

The charging window is what decides it

Fins buy *rate*, so their worth is set by how much rate is demanded:

t_{ch} [h]	bare	24 fins	gain
4	28.976	15.794	1.84
6	22.579	12.499	1.81
10	16.569	11.573	1.43
14	13.468	11.573	1.16
20	11.432	11.573	0.99

Break-even is near 20 h. The origin is visible in the closed form of §4.2: putting $x = r_{\text{cell}}/r_e$ in Eq. (34), a *bare* tube at $\Delta T = 10$ K reaches the cell boundary in

$$t = \frac{\rho_m h_m r_e^2}{k_m \Delta T} \left[\tfrac{1}{2} x^2 \ln x - \tfrac{1}{4} x^2 + \tfrac{1}{4} \right] \approx 6.4\,\text{h}, \tag{83}$$

comfortably inside the 10 h window — which is why a bare tube *nearly* works here, and why the fins look marginal. Halve the window and they are not marginal at all.

7.2 H9 — azimuthal insulation between cascade layers

Assumed perfect in v0.4a so that the rest of the model can be evaluated. This section states what is being assumed away, and what it is worth, so the assumption can be tested rather than inherited.

Where the requirement comes from

The cascade (H8) assigns melting temperatures along the *developed* tube length $L_{\text{tube}} = 2L_{\text{well}}$, because that is the coordinate the fluid actually traverses. A hairpin's down leg occupies $s \in [0, L_{\text{well}}]$ and its return leg $s \in [L_{\text{well}}, L_{\text{tube}}]$, so by Eq. (3) the two legs at a given depth d sit at developed positions $s = d$ and $s = L_{\text{tube}} - d$ — in different layers:

depth [m]	down leg T_m [°C]	return leg T_m [°C]	gap [K]
0	150.0	101.1	**48.9**
400	143.9	107.2	36.7
800	137.8	113.3	24.4
1200	131.7	119.4	12.2
1524	125.6	125.6	0.0

The borehole cross-section must therefore be *partitioned*: at every depth the PCM around the down legs is a different material from the PCM around the return legs, and the two are asked to hold a temperature difference of up to 48.9 K across a gap of a few centimetres. The design is silent on how this partition is built.

The per-leg area bookkeeping is nonetheless correct. $A_{\text{avail}} = V_{\text{well}}/(n_t L_{\text{tube}})$ and $L_{\text{tube}} = 2L_{\text{well}}$, so the divisor is the total *leg* length $2n_t L_{\text{well}}$ and each of the $2n_t = 4$ legs owns exactly a quarter of the PCM cross-section, $4.5411\text{e} - 3\,\text{m}^2$. That is what Eq. (81) encodes, and it is unaffected by how the partition is drawn — a quarter-disc and a half-of-a-half-disc have the same area, hence the same δ_{merge}.

What it is worth

Treating the interface between adjacent cells as a slab of width and path length both $\sim 2r_{\text{cell}}$ gives $q'_{\text{leak}} \sim k_m\,\Delta T$:

	ΔT [K]	q'_{leak} [W/m]
wellhead	48.9	22.0
mid-depth	24.4	11.0
well bottom	0.0	0.0
useful duty q', time-mean over the charge		82.8

Of order 25 % of the useful duty at the wellhead. Even allowing a shape factor two or three times kinder than a slab, this is a first-order term, and its sign is unfavourable in a specific way: the leak flows from the hot layer to the cold layer, which is exactly the temperature separation the cascade is built to create. An uninsulated partition does not merely lose energy — it partially undoes N_{lay}.

Three consequences follow for the work still to be done.

1. N_{lay} **can no longer be swept for free.** More layers means a finer cascade and a better match to the glide, but also a larger layer-to-layer ΔT at the wellhead and hence a larger leak. There is an optimum, and the present model cannot see it because the leak is zero by assumption.
2. **The neighbour relationship is not uniform.** Within one half of the partition a leg's neighbour carries the *same* T_m, so their fronts meeting is a genuine merge (H3, §7.1). Across the partition the neighbour carries a different T_m, so the same geometric event is a heat leak, not a merge. The model treats all four legs identically and adiabatically.
3. **The partition displaces PCM.** Whatever insulates the two halves occupies borehole volume and therefore reduces V_{well}, which by §5.2 raises N_{capacity} directly. The insulation is not free even before its thermal performance is considered.

How to settle it

The same two-dimensional conduction calculation in the cell that §7.1 requires would also deliver this, since both are questions about what happens at a cell's outer boundary. One calculation, on the real cross-section rather than an equivalent annulus, resolves the corner PCM, the reduced conducting azimuth, the fins' reach into the corners, and the cross-partition leak together. Until then H9 is held as an idealisation and the well counts in this document should be read as optimistic.

7.3 Where the closed-form front fails

Formulation A (§4.2) rests on two premises, both false in this application, acting in opposite directions.

Surface mismatch. The fluid gives up heat through the finned perimeter $P_T = 0.4924\,\mathrm{m}$ of Eq. (24); the front absorbs it through the bare perimeter $2\pi r_e = 0.1324\,\mathrm{m}$ assumed by Eq. (35). Factor 3.72. The front therefore *under*-melts.

Retroactive driving temperature. Equation (35) assumes T_{r_e} constant since $t = 0$, and is handed its current value. Measured at mid-well over the charging window, $T_{r_e} - T_m$ runs $0 \to 8.62\,\mathrm{K}$. Applying the final value retroactively *over*-melts by a factor of about 1.56.

The two are separable by setting $n_f = 0$, which makes the surfaces coincide and leaves only the second error:

n_f	k_w [W/m/K]	$\int q'\mathrm{d}t \,/\, \rho_m h_m V_{\mathrm{melt}}$
24	0.45	1.032
24	45	2.021
0	0.45	0.642
0	45	0.642

With the fins removed the imbalance does not vanish — it inverts, and becomes almost independent of k_w. The near-unity entry in the first row is therefore not evidence of a sound model. It is a *triple* cancellation between the surface mismatch, the retroactive T_{r_e}, and the conductivity error below.

Evidence: `verification/verify_fins_off.py`, `verification/verify_frozen_Tre.py`.

7.4 The wall conductivity argument — corrected in v0.3

Correction notice. Versions of this model up to and including the IHTC paper used the wrong value of the wall thermal conductivity during charging. The results reported in this document for v0.3 and later use the corrected value.

k_w carries two physical meanings, tube-wall conductivity in Eq. (19) and fin conductivity in Eq. (21), through a single positional argument. In the charging chain that slot received the PCM liquid conductivity, 0.45 W/m/K, instead of steel at 45 W/m/K — two orders of magnitude low. The error is visible in the original solver signature, whose parameter is named `k_m_l`.

The consequence was not confined to the wall resistance, which is small either way. Through Eq. (21) it collapsed the fin efficiency to $\eta_f \approx 0.33$, throttling the fluid-side conductance to roughly what a bare cylinder can absorb, and so accidentally compensating the surface mismatch above.

This is why correcting k_w *alone* makes the model worse, and why it could only be corrected together with the front formulation.

From v0.3 the correct value is the default (`Case.charge_uses_wall_conductivity = True`). Setting it `False` reproduces the published v0.1 configuration, which is how the verification of §7.8 is performed. Both switches must be set together: the closed-form front *and* the conductivity error. Setting only one gives a model that never existed.

7.5 Quantities defined twice

Melt-layer conductance h_e. Equation (20) is the cylindrical shell used in `compute_T_re`. The small-δ branch of `compute_U_i` instead falls back to the plane-slab form $h_e = k_m/\delta$. Same physical quantity, two formulas, iterated against each other in the fixed-point loop of §3.5. The two agree as $\delta/r_e \to 0$ and diverge as the layer grows.

E_{well}. A *capacity* per well in Eq. (67), a *duty* per well in §5.3. Different keys in the code, one name, and related by the model input $1/(1+\lambda)$ — Eq. (77).

k_w. Tube-wall conductivity in Eq. (19), fin conductivity in Eq. (21), one positional argument (§7.4). This one has already caused a two-order-of-magnitude error that survived to publication.

Laminar Nusselt number. 4.36 (uniform flux) in `well_marched`, 3.66 (uniform temperature) in the legacy march. For a melting boundary the isothermal value is the defensible one.

Charging pressure-drop end states. The charging drop is evaluated between T_{4c} and T_{4a}, a source-loop pair, whereas T_{2c} is the borehole return. The affected property is the water density, so the error is small, but the temperatures are not the ones the flow actually sees.

7.6 Absent terms

Azimuthal conduction between cells. Hypothesis H9, §7.2. The largest single omission: neighbouring legs are up to 48.9 K apart by construction and the term is simply absent from the balance of §4.5.5.

Fluid thermal inertia. Hypothesis H7, quantified in §4.5.5. Dropping $\partial T/\partial t$ leaves an unresolved startup transient of about 38 min per half-cycle, 6 % of the window, though only 1.2 % of the latent inventory is involved.

Formation heat loss. Hypothesis H6. The borehole wall is adiabatic. Over a 10 h cycle this is defensible; over seasonal storage it is not. The geothermal gradient is carried in the configuration and used in no energy balance.

Natural convection in the melt. Hypothesis H4. Conduction-only transport in the liquid PCM under-predicts the melting rate once the layer is established, and does so increasingly as δ grows. Note that this is the one absent term that argues *against* fins rather than for them.

Sensible heat in the PCM. *Supplied in v0.4* — see §4.5. What remains absent is any *radial resolution* of temperature within a phase: each segment carries one lumped T_{pcm}, so superheat appears uniformly across the liquid annulus rather than diffusing outward from the tube.

Buoyancy head. Hypothesis H7. The 1524 m loop carries roughly ± 7.5 bar of static head, asymmetric between charge and discharge, and it is absent from Eq. (63).

Isentropic efficiencies. §2.1. Both cycles are evaluated between prescribed state points, so η_{ORC} and COP are idealised.

Volume change on melting. §4.6. The 6.9 % expansion is not represented.

Melt-front shape around the fins. §4.4. The fin *metal* is excluded from A_{melt} as of v0.3, but the front is still taken as a circular annulus; physically it bulges along the fins.

Front interaction in the corners between legs. Hypothesis H3, treated at length in §7.1. The geometric limit is now tested at the cell radius rather than the borehole wall, and under Formulation C it cannot be exceeded — but it is *saturated* over 99 % of the well, so the annular resistance runs at the limit of its validity and neither the reduced conducting azimuth nor the fins' reach into the corner PCM is represented. The most consequential open item in the model.

λ as an output. §2.3. The storage loss is still an input, at 5 %, while the model computes $\eta_{\mathrm{storage}} = 0.9364$.

7.7 Error handling

Both v0.1 bisections return the bracket bound when they fail to converge and report it as an answer, and the driver cell continues past it. In `thums_core` these raise `ThumsConvergenceError` carrying the last iterate. Similarly, failed property lookups formerly routed into the $\delta \to 0$ branch of Eq. (20), so that a failed solve was reported as *maximum* heat transfer; they now raise `ThumsPropertyError`.

Retired output names follow the same principle. `N_inventory` and `N_heat` raise a `KeyError` carrying an explanation rather than returning a number, and the guard overrides `.get()` as well as `[]`, because a silent `None` from `.get()` is the exact failure the guard exists to prevent (§5.2).

The value $h_e = 10^9$ in that branch is itself correct — a zero-thickness melt layer has zero conduction resistance — notwithstanding an inverted comment in the source and an earlier audit that repeated it.

7.8 Verification status

Regression. `Case(front="closed_form")` reproduces the frozen v0.1 fixture to 4×10^{-11} on every index; the residual is CoolProp and NumPy version drift.

Energy closure. Structural under Formulations B and C, and therefore *not* an independent check (§4.3).

Analytical. Equation (35) verified against its own defining quadrature to a residual of 10^{-15}.

Numerical. The branchwise-exact enthalpy integration of Eq. (54) agrees with a 200000-substep explicit reference to 3×10^{-13}, including across branch crossings.

Grid. Not converged and not quoted as such: $\varepsilon_{\mathrm{PCM}}$ moves -0.3 % from $n_{\mathrm{times}} = 40$ to 320, one-sided, so the default carries a small known bias.

External validation. None. No experimental or independently computed data set has been identified against which any formulation can be checked. This is the principal gap in the verification ladder, and no amount of internal consistency substitutes for it.

7.9 Version history

version	date	change
0.4a	2026-09	No physics. §4.5 rewritten as a teaching treatment of Formulation C. Reporting rule added (§5.2): N_{capacity} is a lower bound and must not be read across a sweep alone. Melt-front limit corrected from the borehole wall to the cell radius, Eq. (81), and H3 proximity now reported (§7.1); the fin count and the charging window follow from it. The segment equation is derived in full (§4.5.5). Hypothesis **H9** (perfect azimuthal insulation between cascade layers) stated explicitly and quantified (§7.2). Sizing vocabulary reduced to the two independent numbers it always had (§5.2): `N_inventory` and `N_heat` are retired and now raise — the first because it was a silent alias for N_{capacity} that had meant a *different* quantity in v0.3, the second because it was one of three labels for the rate criterion.
0.4	2026-09	Melt front carries *enthalpy* rather than melted area (§4.5): superheat above T_m, subcool below it, no clipping, local melt fraction confined to $[0, 1]$ by construction. Branchwise exact time integration replaces explicit Euler, which is unstable once the PCM has finite heat capacity. Sizing moves to a capacity criterion (§5.2). N_{wells} $12.74 \to 11.57$; spread in local melt fraction $1.321 \to 0.078$. The v0.3 lower-bound comparison is corrected (§5.2); it compared inconsistent bases.
0.3	2026-09	Wall conductivity corrected to steel during charging (§7.4); fin metal excluded from the melted PCM area (§4.4).
0.2	2026-08	Melt front reformulated by energy balance (§4.3); melt state carried across the cycle; two-constraint sizing; latent inventory made density-consistent.
0.1	—	IHTC paper. Closed-form Stefan front, single sizing criterion, wall conductivity error. Reproduced exactly by the regression of §7.8.